Pattern formation in growing tissues: regulating domain growth via nutrient-limitations and mechanical stress





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Abstract

This project focuses on the mathematical modelling of pattern formation in growing tissues. The aim is to investigate the influence of domain growth on pattern formation, as well as different mechanisms that may limit tissue size.

First, reaction–diffusion systems with cross-diffusion are studied on fixed and growing domains. After linearisation, domain growth leads to a non-autonomous system, for which classical stability analysis methods developed for fixed domains can no longer be applied directly. An existing approach for a two-species system with a diagonal diffusion matrix is extended to the case of a non-diagonal diffusion matrix, thereby allowing cross-diffusion to be taken into account. A second extension is then developed for a three-species system. These results provide conditions for identifying the growth of spatial perturbations on a growing domain.

Second, two mechanisms that may regulate tissue growth are investigated. The first model introduces oxygen, supplied at the boundaries of the domain and consumed by the cells. The simulations show that oxygen limitation can cause growth to saturate towards a finite length. A mechanochemical model is then developed to couple tissue growth to its mechanical stress state. With free boundaries, mechanical stresses modify the pattern dynamics but are not sufficient, in the simulations considered, to limit tissue size. Introducing rigid walls, however, imposes a finite length and leads to a progressive reorganisation of the pattern after contact with the walls.

This work therefore combines theoretical analysis and numerical simulations to jointly investigate pattern formation and growth regulation in evolving tissues.

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Chapter 1

Introduction

Many spatial patterns can be observed in biological systems, from the coats of certain animals to the shapes of plants or even fingerprints. More generally, the development of an organism requires the emergence of organised spatial structures from initially poorly differentiated states. Understanding the mechanisms underlying the emergence of such structures, as well as their evolution, is an important question in biology. Mathematical modelling makes it possible to study the mechanisms that may generate such structures and the conditions required for their emergence.

In 1952, Alan Turing published an article entitled "The Chemical Basis of Morphogenesis" [1], in which he proposed an explanation for these biological patterns. He suggested that the asymmetry observed in many biological systems, such as the development of an embryo from a symmetric mass of cells, could be explained by the interactions and diffusion of chemical substances. He then proposed a model for these patterns based on reaction-diffusion equations, involving interactions between chemical molecules, called morphogens.

Gierer and Meinhardt developed this concept by proposing local activation coupled with long-range inhibition as a mechanism to explain pattern formation. The idea is to couple a first chemical species, called the activator, which diffuses slowly and promotes its own production, with a second species, called the inhibitor, which diffuses more rapidly and inhibits the production of the activator [2]. Counter-intuitively, starting from a spatially homogeneous equilibrium state, some spatial perturbations can be amplified when diffusion is taken into account. The equilibrium state then becomes unstable, allowing spatial patterns to form spontaneously.

One of the challenges in pattern formation is the robustness of the resulting patterns. Indeed, on a fixed domain, the final pattern can be sensitive to the initial conditions, meaning that different initial perturbations can lead to different final patterns. Previous studies have shown that domain growth can provide a solution to this

problem and thus generate more robust patterns [3]. This property of growing domains motivates their study. Some studies on reaction-diffusion systems on growing domains consider growth that depends directly on the concentrations of the chemical species, thus allowing a coupling between pattern formation and domain evolution [4]. However, taking growth into account makes the stability analysis of the system more complex and therefore requires the development of new tools.

At the same time, biological tissues generally do not grow indefinitely. The question of limiting the size of the growing domain is therefore particularly important and will play a central role in this work.

Several mechanisms can be explored to explain the regulation of tissue size. Tissue growth can, for instance, be limited by the availability of nutrients required for cell proliferation, or by mechanical constraints. Mechanical and mechanochemical models of pattern formation have already been studied on fixed domains, notably by taking into account interactions between cells and the extracellular matrix [5]. In his book *The Mathematics and Mechanics of Biological Growth* [6], Goriely considers the impact of mechanical stress on growth. He develops the idea that the total deformation of the tissue can be decomposed as the product of a growth deformation and an elastic deformation. In this work, we will seek to combine these two aspects by studying pattern formation in a growing tissue whose growth is coupled to its mechanical state.

We will first present a system of reaction-cross-diffusion equations on a fixed domain. The aim of this chapter is to recall the theoretical framework for Turing pattern formation and to present several tools and examples that will be used later. The following chapter considers the reaction-cross-diffusion system again, this time introducing domain growth. The aim is then to develop new tools for linear stability analysis. Since this analysis has already been studied in the literature for a two-species system with a diagonal diffusion matrix [7], we seek to extend it to a two-species system with a non-diagonal diffusion matrix, and then to a three-species reaction-cross-diffusion system. In the same chapter, we then consider the question of tissue size regulation. To do so, we construct a model in which oxygen, which is necessary for cell proliferation, is supplied at the boundaries of the domain and consumed by the cells. We then investigate whether oxygen availability can lead to a saturation of tissue size.

Finally, the last chapter explores a second mechanism for growth regulation, this time of mechanical origin. We introduce tissue mechanics in order to take into account the deformations and stresses generated during growth, and then couple growth to the stress state of the tissue. We investigate to what extent mechanical stresses

and interactions with the environment can limit domain growth, as well as their consequences for pattern evolution.

In this work, we will consider a one-dimensional domain. Vectors and matrices are written in bold. Details of the numerical implementation used throughout this work are provided in Appendix A. The MATLAB codes developed as part of this work are available at the following address: https://github.com/PruneMiller/Code-TFE.

Chapter 2

Reaction-cross-diffusion on a fixed domain

The aim of this work is to study pattern formation in a growing tissue whose growth eventually becomes limited. Before addressing this problem, we start with studying pattern formation on a fixed domain. This chapter presents the existing theoretical framework for pattern formation on a fixed domain and provides a reference case for the developments introduced in the following chapters.

2.1 Equations of the model

We consider two interacting species distributed along a one-dimensional tissue $\Omega = [0, L_0]$. Their concentrations may vary for two reasons: local reactions can produce or remove each species, while spatial transport redistributes them along the tissue. We denote their concentrations by $u(x,t)$ and $v(x,t)$.

$$\frac{\partial u}{\partial t} = \underbrace{\frac{\partial}{\partial x} \cdot \left(d_1(u,v)\frac{\partial u}{\partial x} + d_2(u,v)\frac{\partial v}{\partial x} \right)}_{\text{Diffusion}} + \underbrace{f(u,v)}_{\text{Reaction}}, \tag{2.1}$$

$$\frac{\partial v}{\partial t} = \underbrace{\frac{\partial}{\partial x} \cdot \left(d_3(u,v)\frac{\partial u}{\partial x} + d_4(u,v)\frac{\partial v}{\partial x} \right)}_{\text{Diffusion}} + \underbrace{g(u,v)}_{\text{Reaction}}. \tag{2.2}$$

The functions f and g describe the local reactions, while the coefficients $d_i(u,v)$ represent the diffusion and cross-diffusion coefficients. The diagonal terms d_1 and d_4 describe standard diffusion: the spatial distribution of each species influences its own flux. The off-diagonal terms d_2 and d_3 describe cross-diffusion: a spatial gradient of one species can induce a flux of the other. When $d_2 = d_3 = 0$, the model reduces to a standard reaction-diffusion system.

We apply zero-flux boundary conditions at both ends of the domain:

$$d_1(u,v)\frac{\partial u}{\partial x} + d_2(u,v)\frac{\partial v}{\partial x} = 0, \tag{2.3}$$

$$d_3(u,v)\frac{\partial u}{\partial x} + d_4(u,v)\frac{\partial v}{\partial x} = 0. \tag{2.4}$$

Thus, the species fluxes at the boundaries are equal to zero: there is no exchange between the tissue and its environment.

Initial conditions are prescribed as $u(x,0) = u_0(x)$ and $v(x,0) = v_0(x)$.

2.2 Linear stability analysis

Our objective is to determine whether this system can spontaneously generate a spatial pattern. We therefore start from a spatially homogeneous equilibrium, corresponding to constant concentrations throughout the domain, and study what happens when it is subjected to a small spatial perturbation. If every small perturbation decays, the homogeneous state is stable. Conversely, if some spatial perturbations grow, the initially homogeneous state can develop spatial heterogeneities and eventually form a pattern. We study this using the linear stability analysis for reaction-diffusion systems presented in [8].

Let (u^*, v^*) be a homogeneous steady state. It satisfies:

$$f(u^*, v^*) = 0, \tag{2.5}$$

$$g(u^*, v^*) = 0. \tag{2.6}$$

We perturb the system by writing $u = u^* + \epsilon U(x,t)$ and $v = v^* + \epsilon V(x,t)$ for $|\epsilon| \ll 1$.

Substituting these into (2.1) and (2.2), and retaining only terms of order ϵ, we obtain:

$$\frac{\partial U(x,t)}{\partial t} = d_1(u^*,v^*)\nabla^2 U(x,t) + d_2(u^*,v^*)\nabla^2 V(x,t) + U(x,t)\frac{\partial f}{\partial u}(u^*,v^*) + V(x,t)\frac{\partial f}{\partial v}(u^*,v^*), \tag{2.7}$$

$$\frac{\partial V(x,t)}{\partial t} = d_3(u^*,v^*)\nabla^2 U(x,t) + d_4(u^*,v^*)\nabla^2 V(x,t) + U(x,t)\frac{\partial g}{\partial u}(u^*,v^*) + V(x,t)\frac{\partial g}{\partial v}(u^*,v^*). \tag{2.8}$$

The system can be written in the following matrix form:

$$\frac{\partial \mathbf{W}(x,t)}{\partial t} = \mathbf{D}^*\frac{\partial^2 \mathbf{W}}{\partial x^2} + \mathbf{J}\mathbf{W}, \tag{2.9}$$

where: $\mathbf{W} = \begin{pmatrix} U \\ V \end{pmatrix}$, $\mathbf{D}^* = \begin{pmatrix} d_1(u^*, v^*) & d_2(u^*, v^*) \\ d_3(u^*, v^*) & d_4(u^*, v^*) \end{pmatrix}$ and $\mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial u}(u^*, v^*) & \frac{\partial f}{\partial v}(u^*, v^*) \\ \frac{\partial g}{\partial u}(u^*, v^*) & \frac{\partial g}{\partial v}(u^*, v^*) \end{pmatrix}$.

To determine which spatial structures can grow, we decompose the perturbation $\mathbf{W}$ into elementary spatial modes. Because the domain is an interval with homogeneous Neumann boundary conditions, a natural basis is provided by the eigenfunctions of the Laplacian satisfying these boundary conditions.

$$\mathbf{W}(x,t) = \sum_{k=0}^{+\infty} \psi_k(x) \tilde{\mathbf{W}}_k(t), \tag{2.10}$$

$$\frac{\partial^2 \psi_k}{\partial x^2} = -\rho_k \psi_k(x), \tag{2.11}$$

where the constant spatial eigenvalues $\rho_k = \left(\frac{k\pi}{L_0}\right)^2$ correspond to the standard Laplacian on an interval $[0, L_0]$ with associated eigenfunctions $\psi_k(x) = \cos\left(\frac{k\pi x}{L_0}\right)$.

Inserting this expression of $\mathbf{W}$ into (2.9), we find:

$$\sum_{k=0}^{+\infty} \psi_k(x) \frac{d\tilde{\mathbf{W}}_k(t)}{dt} = \sum_{k=0}^{+\infty} \left[\left(-\mathbf{D}^* \rho_k + \mathbf{J}\right) \psi_k(x) \tilde{\mathbf{W}}_k(t) \right]. \tag{2.12}$$

By uniqueness of the decomposition in the basis $\{\psi_k\}$, we obtain:

$$\frac{d\tilde{\mathbf{W}}_k(t)}{dt} = \left(-\mathbf{D}^* \rho_k + \mathbf{J}\right) \tilde{\mathbf{W}}_k(t). \tag{2.13}$$

Let $\mathbf{B}_k = -\mathbf{D}^* \rho_k + \mathbf{J}$. Each mode k corresponds to a spatial perturbation with wavenumber $\frac{k\pi}{L_0}$. Therefore, studying the stability of each $\mathbf{B}_k$ determines which spatial perturbations can grow and potentially generate a pattern.

Let $\mathbf{q}_k$ be an eigenvector of $\mathbf{B}_k$ associated with the eigenvalue λ_k. We seek solutions of the form: $\tilde{\mathbf{W}}_k(t) = \mathbf{q}_k e^{\lambda_k t}$. Substituting in equation (2.13), we obtain:

$$\lambda_k^{\pm} = \frac{\mathrm{tr}(\mathbf{B}_k) \pm \sqrt{\mathrm{tr}(\mathbf{B}_k)^2 - 4\det(\mathbf{B}_k)}}{2}. \tag{2.14}$$

If $\max(\Re(\lambda_k^+), \Re(\lambda_k^-)) < 0$, the mode k is stable. If $\max(\Re(\lambda_k^+), \Re(\lambda_k^-)) > 0$, the mode k is unstable.

This can be visualised using a dispersion relation, which represents the growth rate of spatial perturbations as a function of their wavelength. An example is shown

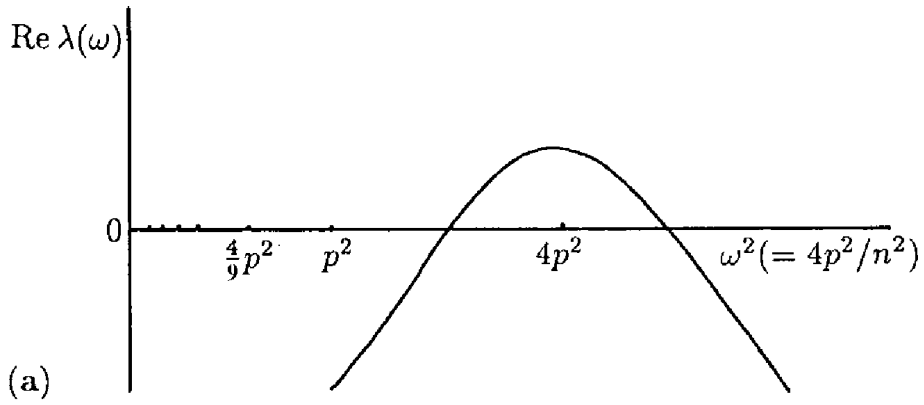


Figure 2.1: Example of a dispersion relation for a reaction-diffusion system. Spatial perturbations associated with wavelengths for which $\mathrm{Re}(\lambda) > 0$ are unstable and can grow. In the notation used in the figure, only the mode $n = 1$ lies in the unstable region. Perturbations for which $\mathrm{Re}(\lambda) < 0$ decay. Reproduced from [8].

in fig. 2.1. Perturbations corresponding to $\mathrm{Re}(\lambda) > 0$ grow, while those corresponding to $\mathrm{Re}(\lambda) < 0$ decay. Therefore, only a range of spatial wavelengths can be unstable.

As $\rho_0 = 0$, $\mathbf{B_0} = \mathbf{J}$ so the mode $k = 0$ is a homogeneous perturbation. The homogeneous steady state is stable if $\max(\Re(\lambda_0^+), \Re(\lambda_0^-)) < 0$, which is equivalent to the conditions $\det(\mathbf{J}) > 0$ and $\mathrm{tr}(\mathbf{J}) < 0$.

For $k \geq 1$, the stability of each mode is determined by the eigenvalues of $\mathbf{B}_k$. A diffusion-driven instability occurs when the homogeneous steady state is stable and at least one mode k is unstable:

$$\max(\Re(\lambda_0^+), \Re(\lambda_0^-)) < 0, \tag{2.15}$$

$$\exists\, k \geq 1 \quad \max(\Re(\lambda_k^+), \Re(\lambda_k^-)) > 0. \tag{2.16}$$

The dominant mode is defined as $k_{\mathrm{dom}} = \arg\max_{k \geq 1}(\max(\Re(\lambda_k^+), \Re(\lambda_k^-)))$.

2.3 Two reference models

2.3.1 Schnakenberg model

As a first example, we consider the Schnakenberg reaction-diffusion model, defined by the following equations:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + f(u, v), \tag{2.17}$$

$$\frac{\partial v}{\partial t} = D\frac{\partial^2 v}{\partial x^2} + g(u,v), \tag{2.18}$$

where:

$$f(u,v) = a + u^2 v - u, \tag{2.19}$$

$$g(u,v) = b - u^2 v, \tag{2.20}$$

with a and b positive parameters.

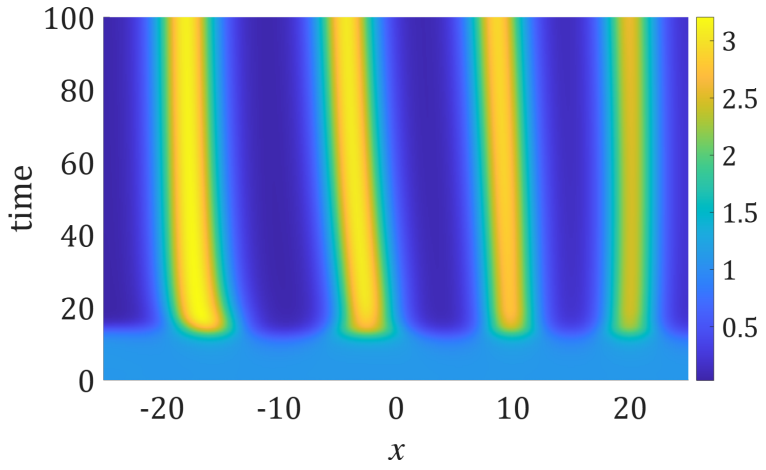


Figure 2.2: Evolution of the cell concentration u for the Schnakenberg model on a fixed domain, with parameters $a = 0.01$, $b = 1.1$, $D = 40$ and $L_0 = 50$.

The homogeneous steady state of this system is

$$\left(a + b, \frac{b}{(a+b)^2}\right). \tag{2.21}$$

The Jacobian is given by:

$$\mathbf{J} = \begin{pmatrix} \frac{b-a}{a+b} & (a+b)^2 \\ -\frac{2b}{a+b} & -(a+b)^2 \end{pmatrix}. \tag{2.22}$$

Then:

$$\mathbf{B}_k = \begin{pmatrix} \frac{b-a}{a+b} - \rho_k & (a+b)^2 \\ -\frac{2b}{a+b} & -(a+b)^2 - D\rho_k \end{pmatrix}. \tag{2.23}$$

The stability of each spatial mode can then be determined from the eigenvalues of $\mathbf{B}_k$, following the general criterion derived above. Figure 2.2 illustrates the emergence of a spatial pattern from a small perturbation of the homogeneous steady state. In this figure, until $t \approx 15$, the concentration remains nearly homogeneous. The perturbations then start to grow, leading to the formation of four peaks of cell concentration. This model will later be used as a reference case to investigate the effect of domain growth on pattern formation.

2.3.2 Keller-Segel model

We then study the Keller-Segel reaction-cross-diffusion model, defined by the following equations:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \cdot \left(\frac{\partial u}{\partial x} - \frac{\chi_0 u}{1+u^2} \frac{\partial v}{\partial x} \right) + f(u,v), \tag{2.24}$$

$$\frac{\partial v}{\partial t} = D \frac{\partial^2 v}{\partial x^2} + g(u,v), \tag{2.25}$$

where:

$$f(u,v) = ru(1-u), \tag{2.26}$$

$$g(u,v) = u - av, \tag{2.27}$$

with D, r, a and χ_0 positive parameters.

The homogeneous steady state of this system is

$$\left(1, \frac{1}{a}\right). \tag{2.28}$$

The Jacobian is given by:

$$\mathbf{J} = \begin{pmatrix} -r & 0 \\ 1 & -a \end{pmatrix}. \tag{2.29}$$

Then:

$$\mathbf{B}_k = \begin{pmatrix} -r - \rho_k & \frac{\chi_0}{2}\rho_k \\ 1 & -a - D\rho_k \end{pmatrix}. \tag{2.30}$$

The trace and the determinant of $\mathbf{B}_k$ are respectively:

$$\mathrm{tr}(\mathbf{B}_k) = -r - a - (1+D)\rho_k < 0, \tag{2.31}$$

$$\det(\mathbf{B}_k) = D\rho_k^2 + \left(rD + a - \frac{\chi_0}{2} \right) \rho_k + ra. \tag{2.32}$$

Hence, the mode k is unstable if and only if $D\rho_k^2 + \left(rD + a - \frac{\chi_0}{2} \right) \rho_k + ra < 0$.

For an instability to occur, this polynomial in ρ_k must be negative for some ρ_k. Since $D > 0$, this requires $rD + a - \frac{\chi_0}{2} < 0$ and $\Delta = \left(rD + a - \frac{\chi_0}{2} \right)^2 - 4Dra > 0$. These two conditions lead to $\chi_0 > 2(\sqrt{rD} + \sqrt{a})^2$.

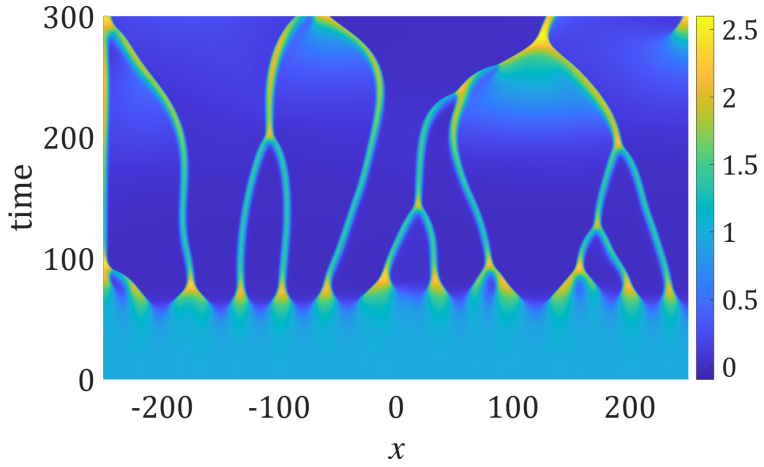


Figure 2.3: Evolution of the cell concentration u for the Keller-Segel model on a fixed domain, with parameters $a = 0.01$, $r = 0.1$, $\chi_0 = 20$, $D = 40$ and $L_0 = 500$.

A simulation of the Keller-Segel model can be found in fig. 2.3.At early times, the concentration is nearly homogeneous. Spatial perturbations then grow and several peaks progressively emerge. Unlike in the Schnakenberg example, the resulting pattern does not remain stationary: the peaks continue to evolve over time, with successive splitting and merging events.

The analysis presented in this chapter provides a general framework for studying reaction-cross-diffusion on a fixed domain. The two examples presented will be used in the other parts, where domain growth makes the linearized system non-autonomous and requires a different stability analysis.

Chapter 3

Reaction-cross-diffusion on a growing domain

On a large fixed domain, several unstable modes can coexist. Since the admissible wavenumbers become more closely spaced as the domain size increases, a larger domain may support a broader range of pattern-forming modes. Since pattern selection can be sensitive to initial conditions, the same system may lead to various final patterns. In some cases, this sensitivity is consistent with biological reality. We can take zebras as an example, as their coats display different patterns from one individual to another.

However, this variability is not always desirable, particularly in segmentation models. To improve robustness in pattern selection, previous work has shown that growth can provide a mechanism to obtain robust pattern formation. Indeed, the initial conditions may influence the first pattern, but since the domain is initially small, few pattern modes are accessible. Then, as the domain grows, the pattern evolves progressively rather than being selected again from all possible modes according to the initial conditions. The existing pattern constrains the transition towards the next pattern [3].

This motivates the study of pattern formation on a growing domain rather than on a fixed one.

I first present the reaction-cross-diffusion model studied on a growing domain. I then perform a linear stability analysis of the resulting non-autonomous system in order to determine under which conditions spatial perturbations can grow and lead to pattern formation. After presenting the existing instability criterion developed by Van Gorder [7] for a two-species system with diagonal diffusion, I extend this approach to a two-species system with a non-diagonal diffusion matrix, allowing cross-diffusion to be taken into account, and then to a three-species system. In the final section

of this chapter, I modify the model to include oxygen supplied at the boundaries of the domain and investigate whether oxygen limitation can prevent indefinite tissue growth.

3.1 Equations of the model

Now that the domain is growing, we introduce a Lagrangian description in which the material coordinate in the reference configuration is denoted by $X \in [0, L_0]$, following the approach used for reaction-diffusion systems on growing domains by [4]. The current position of the material point initially located at X is denoted by $x = x(X, t)$. We can then define the local domain stretch as

$$\mu(X, t) = \frac{\partial x}{\partial X}. \tag{3.1}$$

μ describes the local change in length: thus, $\mu > 1$ corresponds to local expansion, while $\mu < 1$ corresponds to local contraction. Its evolution is given by:

$$\frac{\partial \mu}{\partial t} = \mu S(t, u, v), \tag{3.2}$$

where S is a prescribed function describing the rate of local expansion or contraction.

We then consider a reaction-cross-diffusion system of two species, u and v, on a growing one-dimensional domain:

$$\frac{\partial u}{\partial t} = \underbrace{\frac{1}{\mu(X,t)} \frac{\partial}{\partial X} \left(\frac{d_1(u,v)}{\mu(X,t)} \frac{\partial u}{\partial X} + \frac{d_2(u,v)}{\mu(X,t)} \frac{\partial v}{\partial X} \right)}_{\text{Cross-diffusion}} - \underbrace{uS(t,u,v)}_{\text{Dilution}} + \underbrace{f(u,v)}_{\text{Reaction}}, \tag{3.3}$$

$$\frac{\partial v}{\partial t} = \underbrace{\frac{1}{\mu(X,t)} \frac{\partial}{\partial X} \left(\frac{d_3(u,v)}{\mu(X,t)} \frac{\partial u}{\partial X} + \frac{d_4(u,v)}{\mu(X,t)} \frac{\partial v}{\partial X} \right)}_{\text{Cross-diffusion}} - \underbrace{vS(t,u,v)}_{\text{Dilution}} + \underbrace{g(u,v)}_{\text{Reaction}}. \tag{3.4}$$

The temporal evolution of species densities is governed by three contributions: reaction terms f and g; diffusion and cross-diffusion terms, which now depend on $\mu(X, t)$; a dilution term, which stems from the tissue growth. As the tissue expands, the same amount of material is distributed over a larger region, so the local concentration decreases.

Hence, growth affects the equations in two ways: directly through the dilution term, and indirectly through the modification of the diffusion operator.

We impose zero-flux boundary conditions on both sides:

$$\left(\frac{d_1(u,v)}{\mu(X,t)}\frac{\partial u}{\partial X} + \frac{d_2(u,v)}{\mu(X,t)}\frac{\partial v}{\partial X}\right)\Bigg|_{X=0,L_0} = 0, \tag{3.5}$$

$$\left(\frac{d_3(u,v)}{\mu(X,t)}\frac{\partial u}{\partial X} + \frac{d_4(u,v)}{\mu(X,t)}\frac{\partial v}{\partial X}\right)\Bigg|_{X=0,L_0} = 0. \tag{3.6}$$

3.2 An example: Schnakenberg model

Implementing the Schnakenberg model of section 2.3.1 on a growing domain, we obtain fig. 3.1.

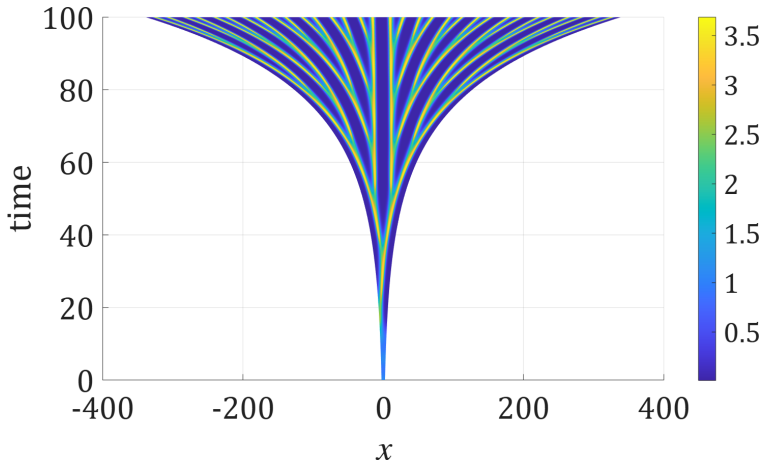


Figure 3.1: Evolution of the cell concentration u for the Schnakenberg model on a growing domain, with parameters $a = 0.01$, $b = 1.1$, $D = 40$, $S(t,u,v) = 0.049082$ and $L_0 = 5$.

At the beginning, until $t \approx 15$, the domain is homogeneous. Then, the domain is still small and only one peak is visible. Then, the pattern evolves as the tissue grows, and new peaks progressively appear. The peaks undergo simultaneous splitting, with each peak splitting into two.

3.3 Linear stability analysis

3.3.1 Homogeneous base state

Let $(u^*(t), v^*(t), \mu^*(t))$ be a homogeneous base state of equations (3.2), (3.3) and (3.4) that evolves as:

$$\frac{\partial u^*}{\partial t} = -u^* S(t,u^*,v^*) + f(u^*,v^*), \tag{3.7}$$

$$\frac{\partial v^*}{\partial t} = -v^* S(t, u^*, v^*) + g(u^*, v^*), \tag{3.8}$$

$$\frac{\partial \mu^*}{\partial t} = \mu^* S(t, u^*, v^*). \tag{3.9}$$

We choose $u^*(0)$ and $v^*(0)$ to satisfy $f(u^*(0), v^*(0)) = 0$ and $g(u^*(0), v^*(0)) = 0$ to agree with the linear stability analysis on fixed domain, and we take $\mu^*(0) = 1$.

3.3.2 Linearization and mode decomposition

Details of the calculations of this section can be found in Appendix F. Let $H^* := H(u^*, v^*)$ where $H \in \{d_1, d_2, d_3, d_4, f, g\}$ and $S^* := S(t, u^*, v^*)$. Following the same method presented in section 2.2, the perturbed system can be written in the form:

$$\frac{d\tilde{\mathbf{W}}_k(t)}{dt} = \mathbf{B}_k(t)\tilde{\mathbf{W}}_k(t), \tag{3.10}$$

where

$$\tilde{\mathbf{W}}_k = \begin{pmatrix} U_k \\ V_k \end{pmatrix} \tag{3.11}$$

and

$$\mathbf{B}_k(t) = \begin{pmatrix} b_{11}(t) & b_{12}(t) \\ b_{21}(t) & b_{22}(t) \end{pmatrix} = -\frac{\mathbf{D}^*(t)}{(\mu^*(t))^2}\rho_k + \mathbf{A}(t), \tag{3.12}$$

with

$$\mathbf{D}^*(t) = \begin{pmatrix} d_1^*(t) & d_2^*(t) \\ d_3^*(t) & d_4^*(t) \end{pmatrix} \tag{3.13}$$

and

$$\mathbf{A}(t) = \begin{pmatrix} \frac{\partial f^*}{\partial u} - u^*\frac{\partial S^*}{\partial u} - S^* & \frac{\partial f^*}{\partial v} - u^*\frac{\partial S^*}{\partial v} \\ \frac{\partial g^*}{\partial u} - v^*\frac{\partial S^*}{\partial u} & \frac{\partial g^*}{\partial v} - v^*\frac{\partial S^*}{\partial v} - S^* \end{pmatrix}. \tag{3.14}$$

On a fixed domain, the matrix governing each spatial mode is constant in time. The stability of a mode can therefore be determined directly from the eigenvalues of this matrix: a positive real part corresponds to an exponentially growing perturbation. However, on a growing domain, the matrix $\mathbf{B}_k(t)$ is time-dependent: the system is non-autonomous and its solutions cannot, in general, be written in the form $\mathbf{q}_k e^{\lambda_k t}$ with constant $\mathbf{q}_k$ and λ_k. Therefore, unlike in the fixed domain case, the instantaneous eigenvalues of $\mathbf{B}_k(t)$ are not sufficient, in general, to determine whether a perturbation grows or decays over a time interval, which motivates the following section.

3.3.3 Instability analysis

This part is dedicated to studying the stability of spatial modes in the non-autonomous system. To overcome the difficulty of determining the stability of a time-dependent linear system, Van Gorder [9] proposed a different approach for two-species reaction-diffusion systems with a diagonal diffusion matrix. Instead of studying the instantaneous eigenvalues of $\mathbf{B}_k(t)$, the two-dimensional first-order modal system is reduced to a scalar second-order non-autonomous differential equation. Comparison arguments can then be used to obtain sufficient conditions for the growth or decay of the spatial modes. The main result on which this approach relies is the following theorem.

Theorem 1 (Van Gorder, Theorem A.1 [7])**.** *Consider the ordinary differential equation*

$$Y''(t) + P(t)Y'(t) + Q(t)Y(t) = 0 \tag{3.15}$$

on a time interval I. Suppose that there exists a positive function $\Phi(t)$ such that

$$Q(t) \leq -\frac{\Phi''(t)}{\Phi(t)} - \frac{\Phi'(t)}{\Phi(t)}P(t), \qquad \forall t \in I. \tag{3.16}$$

Then, this condition is sufficient for the existence of a fundamental solution $Y(t)$ satisfying

$$|Y(t)| \geq \Phi(t), \qquad \forall t \in I. \tag{3.17}$$

In particular, $Y(t)$ has an exponential growth rate on I if

$$Q(t) < 0, \qquad \forall t \in I. \tag{3.18}$$

Conversely, the reverse strict inequality

$$Q(t) > -\frac{\Phi''(t)}{\Phi(t)} - \frac{\Phi'(t)}{\Phi(t)}P(t), \qquad \forall t \in I \tag{3.19}$$

provides a complementary condition for decay [9].

The theorem allows us to prescribe a positive comparison function $\Phi(t)$ and to determine whether the differential equation admits a solution whose amplitude remains at least as large as $\Phi(t)$ over the considered time interval.

In the present context, we will choose $\Phi(t)$ to represent an exponentially growing perturbation. The resulting inequality will then provide a sufficient condition for the growth of a spatial mode.

Thus, Van Gorder establishes a criterion for a two-species diagonal diffusion matrix system. In the next section, I will extend this criterion first for a non-diagonal diffusion matrix, allowing cross-diffusion terms to be included, by using the same procedure as Van Gorder, and then I will extend it for a three-species system.

3.3.3.1 Extension of Van Gorder's instability criterion to a non-diagonal diffusion matrix

In this section, I will first extend Van Gorder's criterion to a non-diagonal diffusion matrix. The objective is to reduce the two-dimensional first-order modal system to a scalar second-order equation to which theorem 1 can be applied. In contrast with Van Gorder's original derivation, the coefficients $b_{ij}(t)$ now include both diffusion and cross-diffusion contributions.

Reduction to a second-order differential equation

For each k = 0, 1, 2, ..., equation (3.10) can be written as the system:

$$\frac{d\tilde{U}_k(t)}{dt} = b_{11}(t)\tilde{U}_k(t) + b_{12}(t)\tilde{V}_k(t), \tag{3.20}$$

$$\frac{d\tilde{V}_k(t)}{dt} = b_{21}(t)\tilde{U}_k(t) + b_{22}(t)\tilde{V}_k(t). \tag{3.21}$$

Assuming that $b_{12}(t) \neq 0$ for all $t \in I$:

$$\tilde{V}_k(t) = \frac{1}{b_{12}(t)} \left(\frac{d\tilde{U}_k(t)}{dt} - b_{11}(t)\tilde{U}_k(t) \right). \tag{3.22}$$

We use (3.22) in (3.21) to obtain a single second-order ODE for $\tilde{U}_k(t)$, finding:

$$\frac{d^2\tilde{U}_k(t)}{dt^2} + P_{U_k}(t)\frac{d\tilde{U}_k(t)}{dt} + Q_{U_k}(t)\tilde{U}_k(t) = 0, \tag{3.23}$$

with:

$$P_{U_k}(t) = -\left(\frac{b'_{12}(t)}{b_{12}(t)} + b_{11}(t) + b_{22}(t) \right) \tag{3.24}$$

and:

$$Q_{U_k}(t) = \frac{b_{11}(t)b'_{12}(t)}{b_{12}(t)} - b'_{11}(t) - b_{12}(t)b_{21}(t) + b_{11}(t)b_{22}(t). \tag{3.25}$$

We have therefore reduced the original two-dimensional first-order system to a scalar second-order equation of the same form as equation 3.15. Theorem 1 can now be applied to $\tilde{U}_k$.

An analogous derivation can be performed by eliminating $\widetilde{U}_k$ instead, provided that $b_{21}(t) \neq 0$, leading to a second-order equation for $\tilde{V}_k$.

Growth criterion

Applying Theorem A.1 of Van Gorder, a sufficient condition for exponential growth of $\tilde{U}_k(t)$ is:

$$Q_{U_k}(t) < 0 \tag{3.26}$$

$$\Leftrightarrow \quad \det(\mathbf{B}_k(t)) < b_{12}(t)\frac{d}{dt}\left(\frac{b_{11}(t)}{b_{12}(t)}\right). \tag{3.27}$$

Similarly, assuming that $b_{21}(t) \neq 0$ for all $t \in I$, a sufficient condition for exponential growth of $\tilde{V}_k(t)$ is:

$$Q_{V_k}(t) < 0 \tag{3.28}$$

$$\Leftrightarrow \quad \det(\mathbf{B}_k(t)) < b_{21}(t)\frac{d}{dt}\left(\frac{b_{22}(t)}{b_{21}(t)}\right). \tag{3.29}$$

Finally, combining these two conditions leads to proposition 3.1.

Proposition 3.1 (Sufficient growth criterion for non-diagonal diffusion)**.** *Consider the k-th spatial mode of the two-species system*

$$\frac{d\tilde{\mathbf{W}}_k}{dt} = \mathbf{B}_k(t)\tilde{\mathbf{W}}_k, \qquad \mathbf{B}_k(t) = \begin{pmatrix} b_{11}(t) & b_{12}(t) \\ b_{21}(t) & b_{22}(t) \end{pmatrix}. \tag{3.30}$$

Assume that $b_{12}(t) \neq 0$ and $b_{21}(t) \neq 0$ for all $t \in I$. A sufficient condition for exponential growth of the k-th spatial mode over I is

$$\det(\mathbf{B_k}(t)) < \max\left\{ b_{12}(t)\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{b_{11}(t)}{b_{12}(t)}\right), b_{21}(t)\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{b_{22}(t)}{b_{21}(t)}\right)\right\}, \qquad \forall t \in I. \tag{3.31}$$

Therefore, the instability criterion derived by Van Gorder can be extended to a reaction-cross-diffusion system, without requiring the diffusion matrix to be diagonal. In particular, the cross-diffusion terms are directly accounted for through the off-diagonal coefficients $b_{12}(t)$ and $b_{21}(t)$ of the matrix $\mathbf{B}_k(t)$. Condition (3.31) thus provides a sufficient condition for the instability of the k-th spatial mode over the time interval I.

The same reduction can also be combined with the complementary decay condition (3.19), so that Van Gorder's stability analysis can similarly be extended to the non-diagonal diffusion case. In the following, however, we focus on the instability criterion (3.31), which is used to identify growing spatial modes.

In Appendix B, we verify that, in the case of a diagonal diffusion matrix, criterion (3.31) reduces to the original criterion derived by Van Gorder.

Identification of the maximal guaranteed growth rate

Criterion (3.31) identifies modes for which growth can be guaranteed, but it does not quantify how rapidly they can grow. To compare unstable modes, we therefore seek the largest exponential growth rate that can be guaranteed by the comparison criterion.

Using Corollary 1 of [9], we seek the largest value of δ verifying the following inequality:

$$\delta^2 + P_{U_k}\delta + Q_{U_k} \leq 0. \tag{3.32}$$

If $P_{U_k}^2 - 4Q_{U_k} \geq 0$ and the upper root is positive, the maximal exponential growth rate guaranteed by the criterion is:

$$\delta_{U_k} = \frac{-P_{U_k} + \sqrt{P_{U_k}^2 - 4Q_{U_k}}}{2}. \tag{3.33}$$

3.3.3.2 Extension of Van Gorder's instability criterion to a three-species system

In the previous section, we extended Van Gorder's criterion to a non-diagonal diffusion matrix for two species. This derivation relies on the fact that a two-dimensional first-order system can be reduced to a scalar second-order equation.

To investigate whether the same approach can be used for a larger system, I now consider three interacting species. In this case, eliminating two of the three variables leads to a scalar third-order non-autonomous differential equation. The objective is then to derive a comparison criterion analogous to the two-species case.

Formulation of the three-species modal system

Hence, we add a third species p to the system, which becomes:

$$\frac{\partial u}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{d_1(u,v,p)}{\mu(X,t)}\frac{\partial u}{\partial X} + \frac{d_2(u,v,p)}{\mu(X,t)}\frac{\partial v}{\partial X} + \frac{d_3(u,v,p)}{\mu(X,t)}\frac{\partial p}{\partial X}\right) - uS(t,u,v,p) + f(u,v,p), \tag{3.34}$$

$$\frac{\partial v}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{d_4(u,v,p)}{\mu(X,t)}\frac{\partial u}{\partial X} + \frac{d_5(u,v,p)}{\mu(X,t)}\frac{\partial v}{\partial X} + \frac{d_6(u,v,p)}{\mu(X,t)}\frac{\partial p}{\partial X}\right) - vS(t,u,v,p) + g(u,v,p), \tag{3.35}$$

$$\frac{\partial p}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{d_7(u,v,p)}{\mu(X,t)}\frac{\partial u}{\partial X} + \frac{d_8(u,v,p)}{\mu(X,t)}\frac{\partial v}{\partial X} + \frac{d_9(u,v,p)}{\mu(X,t)}\frac{\partial p}{\partial X}\right) - pS(t,u,v,p) + h(u,v,p). \tag{3.36}$$

Following the same method presented in section 2.2, the perturbed system can be written in the form:

$$\frac{d\tilde{\mathbf{W}}_k(t)}{dt} = \left(-\frac{\mathbf{D}^*(t)}{(\mu^*(t))^2}\rho_k + \mathbf{A}(t)\right)\tilde{\mathbf{W}}_k(t), \tag{3.37}$$

where:

$$\mathbf{W} = \begin{pmatrix} U \\ V \\ P \end{pmatrix}, \quad \mathbf{D}^* = \begin{pmatrix} d_1^* & d_2^* & d_3^* \\ d_4^* & d_5^* & d_6^* \\ d_7^* & d_8^* & d_9^* \end{pmatrix}$$

and

$$\mathbf{A}(t) = \begin{pmatrix} \frac{\partial f^*}{\partial u} - u^*\frac{\partial S^*}{\partial u} - S^* & \frac{\partial f^*}{\partial v} - u^*\frac{\partial S^*}{\partial v} & \frac{\partial f^*}{\partial p} - u^*\frac{\partial S^*}{\partial p} \\ \frac{\partial g^*}{\partial u} - v^*\frac{\partial S^*}{\partial u} & \frac{\partial g^*}{\partial v} - v^*\frac{\partial S^*}{\partial v} - S^* & \frac{\partial g^*}{\partial p} - v^*\frac{\partial S^*}{\partial p} \\ \frac{\partial h^*}{\partial u} - p^*\frac{\partial S^*}{\partial u} & \frac{\partial h^*}{\partial v} - p^*\frac{\partial S^*}{\partial v} & \frac{\partial h^*}{\partial p} - p^*\frac{\partial S^*}{\partial p} - S^* \end{pmatrix}.$$

Let $\mathbf{B}_k(t) = \begin{pmatrix} b_{11}(t) & b_{12}(t) & b_{13}(t) \\ b_{21}(t) & b_{22}(t) & b_{23}(t) \\ b_{31}(t) & b_{32}(t) & b_{33}(t) \end{pmatrix} = -\frac{\mathbf{D}^*(t)}{(\mu^*(t))^2}\rho_k + \mathbf{A}(t)$.

Reduction to a third-order differential equation

For the two-species system, the approach was based on the reduction of the two-dimensional first-order modal system to a scalar second-order non-autonomous ordinary differential equation. For the three-species system, we now consider a three-dimensional first-order modal system, which we reduce to a scalar third-order non-autonomous ordinary differential equation.

For each k = 0, 1, 2, ..., equation (3.37) can be written as the system:

$$\frac{d\tilde{U}_k(t)}{dt} = b_{11}(t)\tilde{U}_k(t) + b_{12}(t)\tilde{V}_k(t) + b_{13}(t)\tilde{P}_k(t), \tag{3.38}$$

$$\frac{d\tilde{V}_k(t)}{dt} = b_{21}(t)\tilde{U}_k(t) + b_{22}(t)\tilde{V}_k(t) + b_{23}(t)\tilde{P}_k(t), \tag{3.39}$$

$$\frac{d\tilde{P}_k(t)}{dt} = b_{31}(t)\tilde{U}_k(t) + b_{32}(t)\tilde{V}_k(t) + b_{33}(t)\tilde{P}_k(t). \tag{3.40}$$

Starting from the first equation of the modal system and differentiating it twice with respect to time, the variables $\tilde{V}_k$ and $\tilde{P}_k$ can be eliminated. Provided that $\Delta = \beta_3 b_{12}(t) - \beta_2 b_{13}(t) \neq 0$, this procedure leads to a scalar third-order non-autonomous differential equation for $\tilde{U}_k$:

$$\tilde{U}_k'''(t) + P_{U_k}(t)\tilde{U}_k''(t) + Q_{U_k}(t)\tilde{U}_k'(t) + R_{U_k}(t)\tilde{U}_k(t) = 0, \tag{3.41}$$

with:

$$P_{U_k}(t) = \frac{\gamma_2}{\Delta} b_{13} - \frac{\gamma_3}{\Delta} b_{12}, \tag{3.42}$$

$$Q_{U_k}(t) = \frac{\gamma_3}{\Delta} \beta_2 - \frac{\gamma_2}{\Delta} \beta_3, \tag{3.43}$$

and:

$$R_{U_k}(t) = -\gamma_1 + \frac{\gamma_2}{\Delta}(\beta_3 b_{11} - b_{13}\beta_1) + \frac{\gamma_3}{\Delta}(b_{12}\beta_1 - \beta_2 b_{11}). \tag{3.44}$$

The detailed elimination procedure and the definitions of the intermediate coefficients β_i and γ_i are given in Appendix G.

We have therefore reduced the original three-dimensional first-order system to a scalar third-order equation. However, Theorem 1 can no longer be applied to $\tilde{U}_k$.

Sufficient growth criterion

The goal is to seek a solution of (**??**) for which there exists a function $\phi(t) > 0$ such that $\tilde{U}_k \geq \phi(t)$. To this end, we set $\tilde{U}_k(t) = \phi(t) Y_k(t)$ and aim to show that $Y_k(t) \geq 1$.

Then:

$$\tilde{U}_k'(t) = \phi'(t) Y_k(t) + \phi(t) Y_k'(t), \tag{3.45}$$

$$\tilde{U}_k''(t) = \phi''(t) Y_k(t) + 2\phi'(t) Y_k'(t) + \phi(t) Y_k''(t), \tag{3.46}$$

$$\tilde{U}_k'''(t) = \phi'''(t) Y_k(t) + 3\phi''(t) Y_k'(t) + 3\phi'(t) Y_k''(t) + \phi(t) Y_k'''(t). \tag{3.47}$$

Substituting (3.45) - (3.47) in (3.41), we obtain:

$$Y_k'''(t) + \zeta_1 Y_k''(t) + \zeta_2 Y_k'(t) + \zeta_3 Y_k(t) = 0, \tag{3.48}$$

where

$$\zeta_1 = 3\frac{\phi'(t)}{\phi(t)} + P_{U_k}(t), \tag{3.49}$$

$$\zeta_2 = 3\frac{\phi''(t)}{\phi(t)} + 2P_{U_k}(t)\frac{\phi'(t)}{\phi(t)} + Q_{U_k}(t), \tag{3.50}$$

and

$$\zeta_3 = \frac{\phi'''(t)}{\phi(t)} + P_{U_k}(t)\frac{\phi''(t)}{\phi(t)} + Q_{U_k}(t)\frac{\phi'(t)}{\phi(t)} + R_{U_k}(t). \tag{3.51}$$

Let $\mathbf{X} = \begin{pmatrix} Y_k \\ Y_k' \\ Y_k'' \end{pmatrix}$.

Equation (3.48) can be written as:

$$\mathbf{X}' = \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\zeta_3(t) & -\zeta_2(t) & -\zeta_1(t) \end{pmatrix}}_{\mathbf{M}_k(t)} \mathbf{X}. \tag{3.52}$$

Assume that $\zeta_2(t) \leq 0$ and $\zeta_3(t) \leq 0$ for all $t \in I$. Then, all the off-diagonal entries of $\mathbf{M}_k(t)$ are non-negative. The system is said to be cooperative. Therefore, if $\mathbf{X}(t_0) \geq 0$, the solution satisfies $\mathbf{X}(t) \geq 0$ for all $t \in I$ [10].

By choosing the initial conditions $\tilde{U}_k(t_0) = \phi(t_0)$, $\tilde{U}_k'(t_0) = \phi'(t_0)$ and $\tilde{U}_k''(t_0) = \phi''(t_0)$, we obtain $Y_k(t_0) = 1$, $Y_k'(t_0) = 0$ and $Y_k''(t_0) = 0$.

Thus, $\mathbf{X}(t_0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \geq 0$.

By the cooperativity of the system, $\mathbf{X}(t) \geq 0$. In particular, $Y_k'(t) \geq 0$, so $Y_k(t)$ is increasing. Then, $Y_k(t) \geq Y_k(t_0) = 1$.

Consequently, $\tilde{U}_k(t) = \phi(t)Y_k(t) \geq \phi(t)$.

To conclude, the result can be summarized by proposition 3.2.

Proposition 3.2 (Sufficient growth criterion for a three-species system)**.** *Consider the k-th mode of the three-species system and assume that it can be reduced to the scalar third-order equation*

$$\tilde{U}_k''' + P_{U_k}(t)\tilde{U}_k'' + Q_{U_k}(t)\tilde{U}_k' + R_{U_k}(t)\tilde{U}_k = 0. \tag{3.53}$$

Let $\phi(t) > 0$ be a prescribed function. Assume that, for all $t \in I_k$,

$$\beta_3(t)b_{12}(t) - \beta_2(t)b_{13}(t) \neq 0, \tag{3.54}$$

and

$$3\frac{\phi''(t)}{\phi(t)} + 2P_{U_k}(t)\frac{\phi'(t)}{\phi(t)} + Q_{U_k}(t) \leq 0, \tag{3.55}$$

$$\frac{\phi'''(t)}{\phi(t)} + P_{U_k}(t)\frac{\phi''(t)}{\phi(t)} + Q_{U_k}(t)\frac{\phi'(t)}{\phi(t)} + R_{U_k}(t) \leq 0. \tag{3.56}$$

Then there exists a solution of the third-order equation satisfying

$$\tilde{U}_k(t) \geq \phi(t), \qquad \forall t \in I_k. \tag{3.57}$$

Hence, if $\phi(t)$ is increasing, these conditions provide a sufficient criterion for the growth of the k-th spatial mode.

Exponential growth criterion

To obtain a more explicit criterion, we now follow the approach used by Van Gorder and consider a specific choice of the comparison function ϕ. We consider $\phi(t) = e^{\delta t}$ for some $\delta > 0$. Inequalities (3.55) and (3.56) become:

$$3\delta^2 + 2P_{U_k}\delta + Q_{U_k} \leq 0, \tag{3.58}$$

$$\delta^3 + P_{U_k}\delta^2 + Q_{U_k}\delta + R_{U_k} \leq 0. \tag{3.59}$$

Equations (3.58) and (3.59) can be written as:

$$\frac{\partial f_k}{\partial \delta}(t, \delta) \leq 0, \tag{3.60}$$

$$f_k(t, \delta) \leq 0, \tag{3.61}$$

for all $t \in I_k$, where $f_k(t, \delta) = \delta^3 + P_{U_k}\delta^2 + Q_{U_k}\delta + R_{U_k}$.

For all $t \in I_k$, inequality (3.60) has real solutions if and only if:

$$\forall t \in I_k, P_{U_k}(t)^2 - 3Q_{U_k}(t) \geq 0. \tag{3.62}$$

In this case, the inequality is verified for all $\delta \in \left[\frac{-P_{U_k} - \sqrt{P_{U_k}^2 - 3Q_{U_k}}}{3}, \frac{-P_{U_k} + \sqrt{P_{U_k}^2 - 3Q_{U_k}}}{3}\right]$.

We note

$$\mathcal{S}_k = \left(\bigcap_{t \in I_k} \left[\frac{-P_{U_k} - \sqrt{P_{U_k}^2 - 3Q_{U_k}}}{3}, \frac{-P_{U_k} + \sqrt{P_{U_k}^2 - 3Q_{U_k}}}{3}\right]\right) \cap]0, +\infty[.$$

From inequality (3.60) we have that $\delta \mapsto f_k(t, \delta)$ decreases. Thus, assuming that $\mathcal{S}_k \neq \emptyset$, if

$$\max_{t \in I_k} f_k \left(t, \min_{t \in I_k} \left(\frac{-P_{U_k}(t) + \sqrt{P_{U_k}(t)^2 - 3Q_{U_k}(t)}}{3} \right) \right) \leq 0, \tag{3.63}$$

then $\delta = \min_{t \in I_k} \left(\frac{-P_{U_k}(t) + \sqrt{P_{U_k}(t)^2 - 3Q_{U_k}(t)}}{3} \right)$ satisfies both inequalities. If not, no $\delta \in \mathcal{S}_k$ satisfies both inequalities for all $t \in I_k$.

Remark 1. To obtain a sufficient criterion that does not depend explicitly on δ, we can consider the limit $\delta \to 0^+$. We obtain the following strict inequalities for all $t \in I_k$:

$$Q_{U_k} < 0, \tag{3.64}$$

$$R_{U_k} < 0. \tag{3.65}$$

If both strict inequalities are verified, assuming that I_k is compact and that the coefficients are continuous, then by continuity, there is a $\delta > 0$ which satisfies (3.58) and (3.59). Thus, if (3.64) and (3.65) are verified, then the mode grows exponentially.

If these conditions are not satisfied, exponential growth may still occur for a finite value of $\delta > 0$, but this criterion is too restrictive to detect it. In this case, the original inequalities (3.58) and (3.59) can be considered directly. ◇

Remark 2. Contrary to the non-diagonal two-species case, Van Gorder's complementary criterion cannot be directly generalised to a three-species system. Therefore, the criterion derived above is only shown to be sufficient for the growth of the mode, and not necessary. ◇

3.4 Results for the instability regions

3.4.1 Numerical implementation

First, we solve the homogeneous state (u^*, v^*, μ^*) over time.

Then, to determine the instability region detected by our criteria, we construct $\mathbf{B}_k(t) = -\frac{\mathbf{D}^*(t)}{(\mu^*(t))^2}\rho_k + \mathbf{A}(t)$ for each time t and each mode k. For the two-species system, we use the inequality (3.31), and for the three-species system, we use the inequalities (3.58) and (3.59).

For each mode satisfying the criterion, we also determine the largest value of $\delta > 0$ for which the growth condition remains satisfied. This provides the maximal growth rate guaranteed by the criterion and allows us to identify the theoretically maximally growing mode.

Finally, the theoretical predictions are compared with the numerical solution of the full PDE system, which is discretised in space using finite differences and integrated in time using ode15s. The dominant spatial mode of the numerical solution is identified using a Fourier analysis and compared with the theoretically predicted mode.

A numerical validation of the implementation can be found in Appendix C.

3.4.2 Instability regions for a Keller-Segel model

To illustrate the two-species instability criterion with a non-diagonal diffusion matrix, we use the following Keller-Segel model:

$$\frac{\partial u}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{D_u}{\mu(X,t)}\frac{\partial u}{\partial X} - \frac{1}{\mu(X,t)}\frac{\chi_u u}{1+u^2}\frac{\partial v}{\partial X}\right) - uS(t,u,v) + f(t,u,v), \tag{3.66}$$

$$\frac{\partial v}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{D_v}{\mu(X,t)}\frac{\partial v}{\partial X}\right) - vS(t,u,v) + g(t,u,v), \tag{3.67}$$

with:

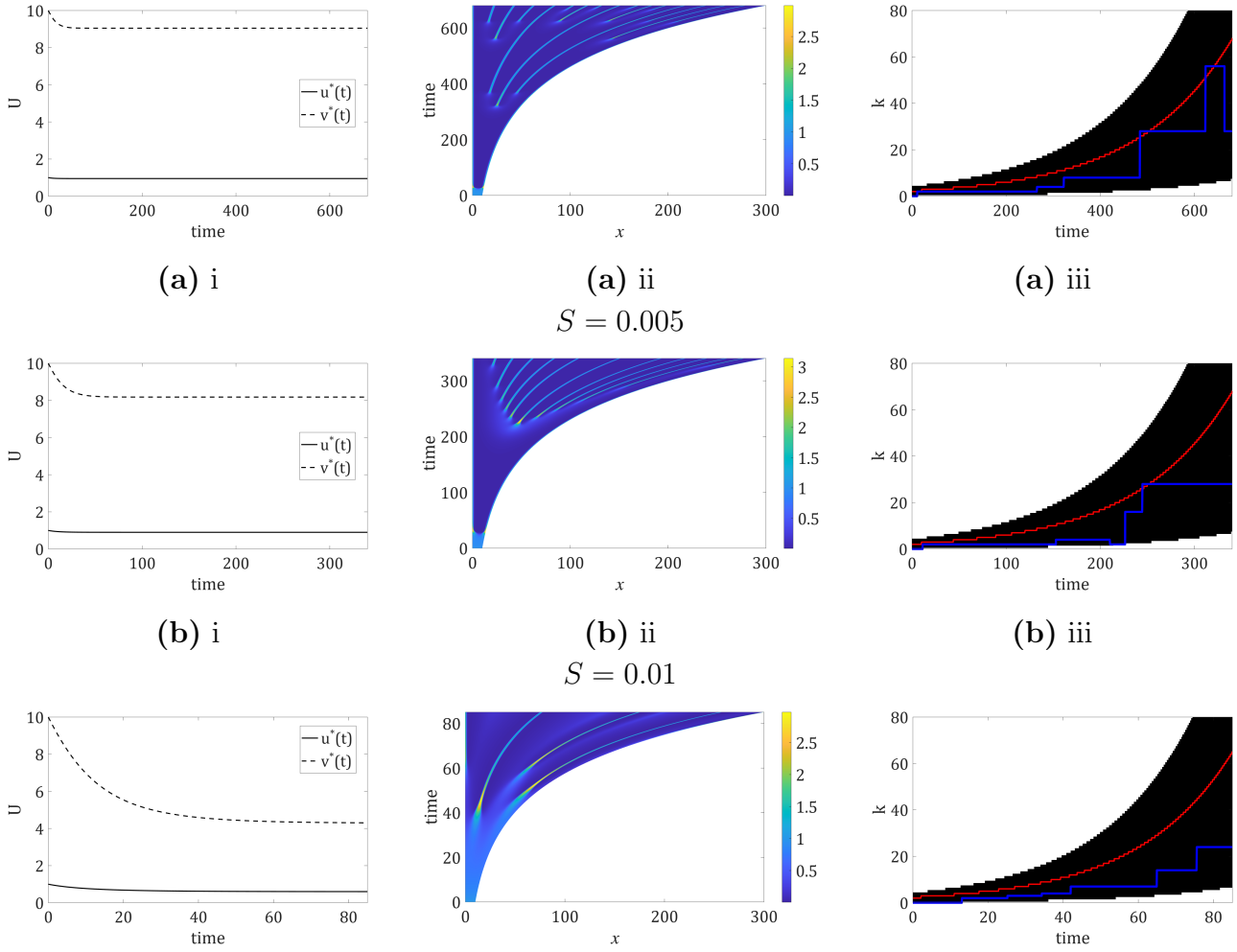
$$f(t,u,v) = ru(1-u), \tag{3.68}$$

$$g(t,u,v) = u - av, \tag{3.69}$$

$$S(t,u,v) = s, \tag{3.70}$$

where $r, a, s > 0$. The results are shown in fig. 3.2. Since the final time is chosen such that the domain reaches the same final size in all simulations, the physical time interval differs between growth rates.

The numerical results show that the theoretical instability criterion provides qualitative information about the modes observed in the full numerical solution, although it does not precisely predict the dominant mode. For slow growth, the pattern has sufficient time to adapt to the evolution of the unstable modes, and successive changes in the dominant numerical mode are observed. For faster growth, the domain evolves more rapidly and the numerical solution remains dominated by lower modes, while the mode with the largest predicted growth rate continues to increase.



(a) i **(a)** ii **(a)** iii

$S = 0.005$

(b) i **(b)** ii **(b)** iii

$S = 0.01$

(c) i **(c)** ii **(c)** iii

$S = 0.04$

Figure 3.2: Simulations for 2 species with parameters $a = 0.1$, $r = 0.1$, $D_u = 1$, $D_v = 1$, $\chi_u = 4$ and $L_0 = 10$. In all simulations, the final time is taken such that the domain has grown to 30 times its initial size. Column (i) shows the homogeneous base state over time. Column (ii) shows the PDE solution u over space and time. Column (iii) shows the set of modes satisfying the instability criterion in black, the mode with the largest predicted growth rate in red, and the largest frequency component of the FFT of $u(x, t)$ from the full numerical solution in blue.

3.4.3 Instability regions for a three-species model

To illustrate the three-species instability criterion, we use the following model:

$$\frac{\partial u}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{D_u}{\mu(X,t)}\frac{\partial u}{\partial X} - \frac{1}{\mu(X,t)}\frac{\chi_u u}{1+u^2}\frac{\partial w}{\partial X}\right) - uS(t,u,v,w) + f(t,u,v,w), \tag{3.71}$$

$$\frac{\partial v}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{D_v}{\mu(X,t)}\frac{\partial v}{\partial X} - \frac{1}{\mu(X,t)}\frac{\chi_v v}{1+v^2}\frac{\partial w}{\partial X}\right) - vS(t,u,v,w) + g(t,u,v,w), \tag{3.72}$$

$$\frac{\partial w}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{D_w}{\mu(X,t)}\frac{\partial w}{\partial X}\right) - wS(t,u,v,w) + h(t,u,v,w), \tag{3.73}$$

with:

$$f(t,u,v,w) = u(1-u), \tag{3.74}$$

$$g(t,u,v,w) = v(1-v), \tag{3.75}$$

$$h(t,u,v,w) = u + v - aw, \tag{3.76}$$

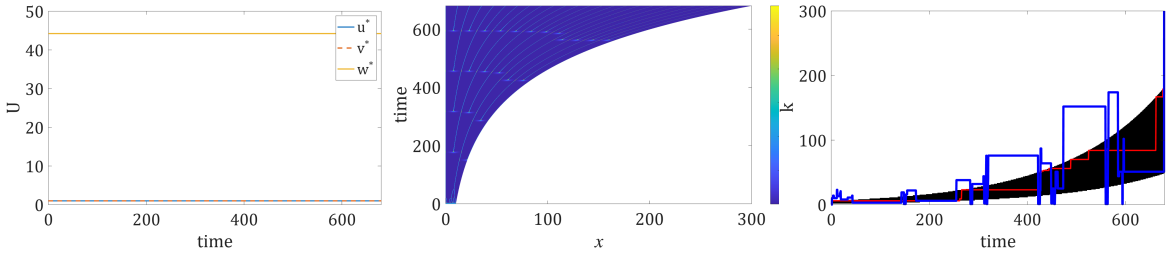
$$S(t,u,v,w) = s, \tag{3.77}$$

where $a, s > 0$.

Results are shown in fig. 3.3.

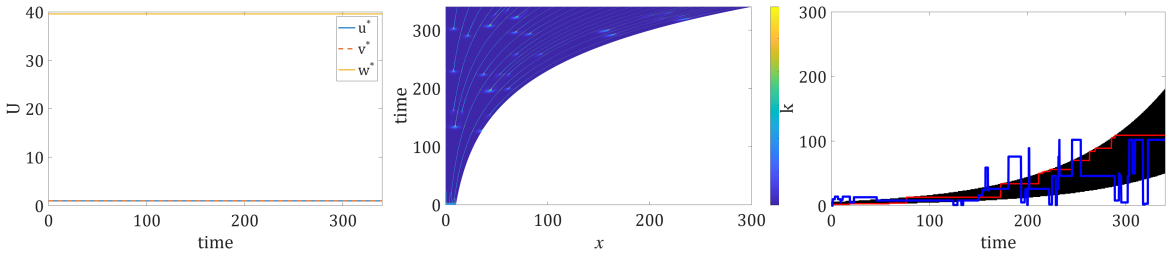
As in the two-species case, the numerical dominant mode generally evolves towards higher spatial modes as the domain grows. However, the agreement between the theoretical prediction and the full numerical solution is less precise for the three-species system. In some cases, the dominant numerical mode lies outside the black region detected by the instability criterion. This does not contradict the theoretical result, since the three-species criterion is only sufficient: a mode outside the black region is not necessarily stable, but simply not detected as growing by the criterion. Similarly, the red curve represents the mode with the largest growth rate guaranteed by the criterion, and therefore does not necessarily correspond to the actual dominant mode of the full system.

For slower growth, successive changes in the dominant numerical mode are observed. For faster growth, fewer mode transitions are observed and the numerical solution does not follow this evolution as closely. This may be explained by the fact that the domain evolves more rapidly relative to the timescale required for unstable modes to develop.



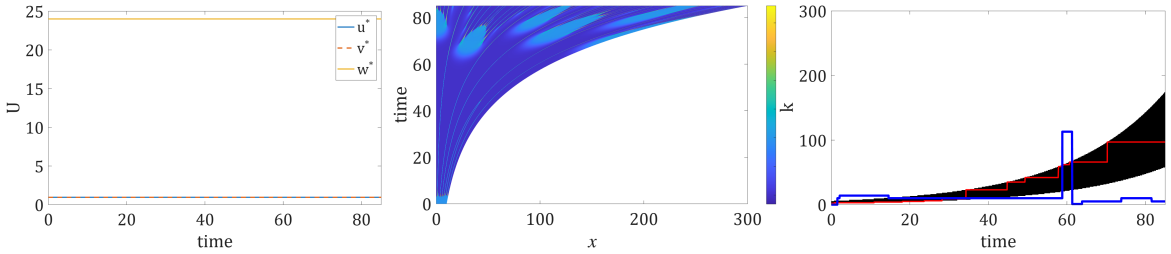
(a) i **(a)** ii **(a)** iii

S = 0.005



(b) i **(b)** ii **(b)** iii

S = 0.01



(c) i **(c)** ii **(c)** iii

S = 0.04

Figure 3.3: Simulations for 3 species with parameters a = 0.04, $D_u = 1$, $D_v = 1$, $D_w = 0.1$, $\chi_u = 6$, $\chi_v = 8$ and $L_0 = 10$. In all simulations, the final time is taken such that the domain has grown to 30 times its initial size. Column (i) shows solutions of the homogeneous base state over time. Column (ii) shows the PDE solution u over space and time. Column (iii) shows the set of modes satisfying the sufficient three-species instability criterion in black, with the mode with the largest growth rate predicted by this criterion in red and the largest frequency component of the FFT of u(x, t) from the full numerical solution in blue.

3.5 Extension of the model to include oxygen

In the previous sections, we studied a reaction-cross-diffusion system on a growing domain, which can increase in size indefinitely. However, biological tissues generally have a finite length, which may be limited by their environment. This motivates the introduction of mechanisms that can regulate domain growth and lead to a finite tissue size.

As a first extension of the reaction-cross-diffusion model, I will study nutrient limitation in this section. In particular, I introduce oxygen into the model, so that its availability can affect cell behaviour and ultimately regulate tissue growth. In the next chapter, I will study a second mechanism for limiting growth, which is mechanical regulation. As a tissue grows and interacts with its mechanical environment, stresses can develop, and these stresses can in turn feed back on the growth rate.

3.5.1 Model

We extend the previous model by including oxygen. Oxygen affects cell behaviour in three ways. First, it acts as a chemoattractant: cells migrate towards high oxygen level regions. Second, high oxygen concentrations stimulate cell proliferation. Finally, cells consume oxygen, leading to spatial variations in oxygen concentration which in turn influence cell migration and proliferation.

The system is governed by the following equations:

$$\frac{\partial c}{\partial t} = \underbrace{\frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{D_c}{\mu(X,t)}\frac{\partial c}{\partial X} - \frac{\chi c}{\mu(X,t)}\frac{\partial o}{\partial X}\right)}_{\text{Diffusion + chemotaxis}} - \underbrace{cS(t,c,o)}_{\substack{\text{Dilution due to}\\ \text{tissue growth}}} + \underbrace{c(H(o)-c)}_{\substack{\text{Proliferation and}\\ \text{death of cells}}}, \quad (3.78)$$

$$\frac{\partial o}{\partial t} = \underbrace{\frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\left(\frac{D_o}{\mu(X,t)}\frac{\partial o}{\partial X}\right)}_{\text{Diffusion}} - \underbrace{oS(t,c,o)}_{\substack{\text{Dilution due to}\\ \text{tissue growth}}} - \underbrace{\lambda co}_{\substack{\text{Consumption of}\\ \text{oxygen by cells}}}, \quad (3.79)$$

$$\frac{\partial \mu}{\partial t} = \mu S(t,c,o), \quad (3.80)$$

where:

$$H(o) = \frac{1}{2}\left[1 + \tanh\left(\alpha\left(o - \frac{o_c}{2}\right)\right)\right] \quad (3.81)$$

and:

$$S(t,c,o) = c - \frac{1}{2}. \quad (3.82)$$

The function H is a smooth Heaviside function, modeling the oxygen availability. It varies between 0 and 1 and is centered at $\frac{o_c}{2}$. Well-oxygenated regions stimulate cell proliferation, which is reduced in poorly oxygenated regions.

The local tissue growth rate S is assumed to depend on the cell density. The threshold $c = \frac{1}{2}$ separates local tissue growth from shrinkage. So in this model, oxygen does not directly control the growth rate.

At the boundaries, we impose Neumann conditions for cells:

$$\frac{D_c}{\mu(0,t)}\frac{\partial c}{\partial X}(0,t) - \frac{\chi c(0,t)}{\mu(0,t)}\frac{\partial o}{\partial X}(0,t) = \frac{D_c}{\mu(L_0,t)}\frac{\partial c}{\partial X}(L_0,t) - \frac{\chi c(L_0,t)}{\mu(L_0,t)}\frac{\partial o}{\partial X}(L_0,t) = 0. \tag{3.83}$$

For oxygen, however, we impose Dirichlet boundary conditions, with a fixed oxygen concentration o_c at both ends of the domain. This represents oxygen being continuously supplied by the external environment:

$$o(0,t) = o(L_0,t) = o_c. \tag{3.84}$$

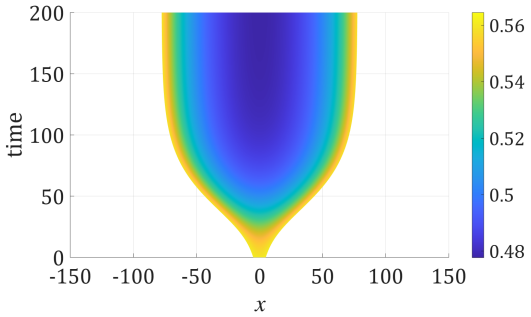
This gives us a parameter that we can control: by changing the value of o_c, we change the amount of oxygen available to the tissue at the boundaries.

3.5.2 Simulations

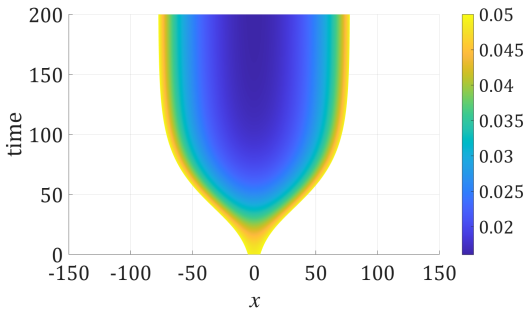
Results are shown in fig. 3.4.

With this growth law, the tissue length reaches a plateau. Initially, the tissue is small enough for oxygen to diffuse throughout the entire domain, leading to its growth. As the distance between the center and the boundaries of the tissue increases, the centre becomes progressively less oxygenated. This drop in oxygen levels affects the tissue through two mechanisms. First, the chemotaxis term drives cell migration from low-oxygen regions toward high-oxygen regions. Second, cell proliferation, modeled by the reaction term $cH(o)$, decreases as o decreases. Consequently, the cell density c drops; since $S(t,c) = c - \frac{1}{2}$, S may become negative, causing the tissue to shrink. This phenomenon is observed at the center. While the center of the tissue undergoes shrinkage, the boundaries, where oxygen levels are highest, continue to grow. Eventually, the shrinkage at the center balances the growth at the edges, and the length of the domain stabilizes. The total domain length therefore reaches a finite plateau, even though the tissue is still undergoing local growth and shrinkage.

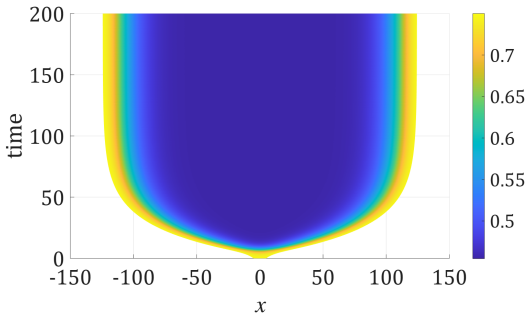
The oxygen level at the boundaries influences the rate at which the domain reaches its final size, as well as its maximum size. The evolution of the domain length over time for different values of o_c is plotted in fig. 3.5.



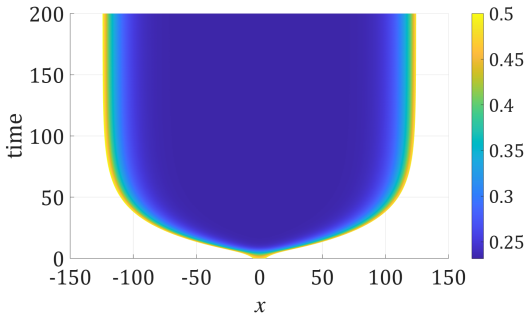
(a) Evolution of c with Neumann boundary conditions and $o_c = 0.05$



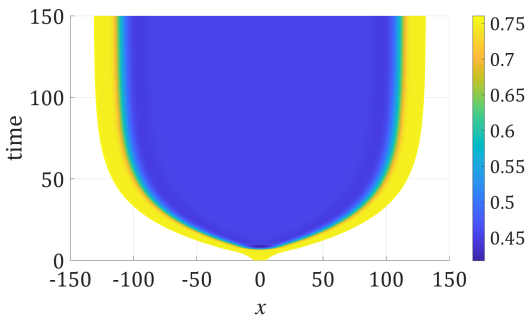
(b) Evolution of o with Dirichlet boundary conditions and $o_c = 0.05$



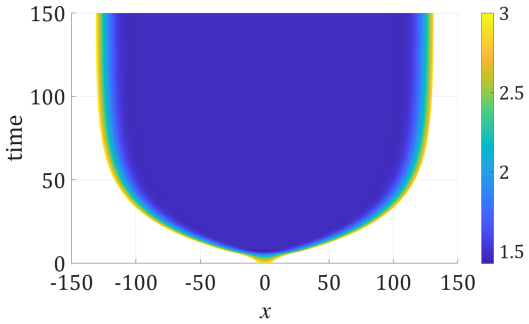
(c) Evolution of c with Neumann boundary conditions and $o_c = 0.5$



(d) Evolution of o with Dirichlet boundary conditions and $o_c = 0.5$

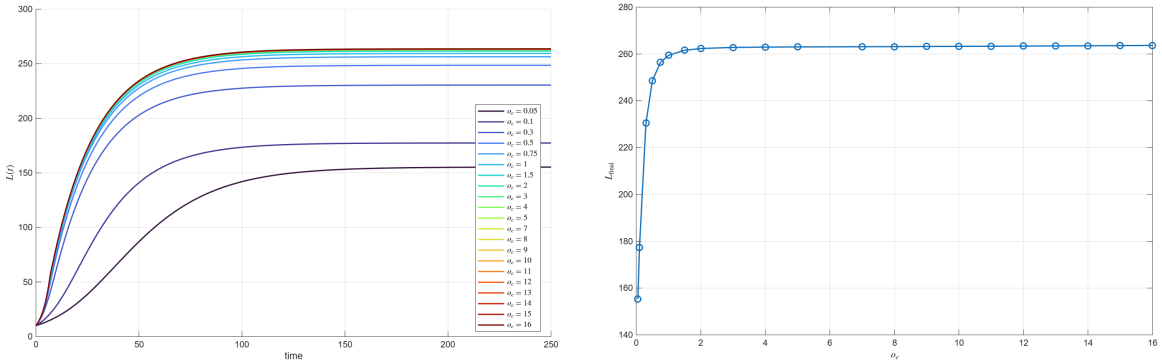


(e) Evolution of c with Neumann boundary conditions and $o_c = 3$



(f) Evolution of o with Dirichlet boundary conditions and $o_c = 3$

Figure 3.4: 1D simulations for one cell population coupled with oxygen under different values of oxygen concentration at the boundaries. In all simulations, the initial domain length is $L_0 = 10$, with $D_c = 1$, $D_o = 60$, $\lambda = 0.1$, $\chi = 0.1$ and $\alpha = 10$. The growth law is given by $S(t, c) = c - \frac{1}{2}$.



(a) Temporal evolution of the domain length

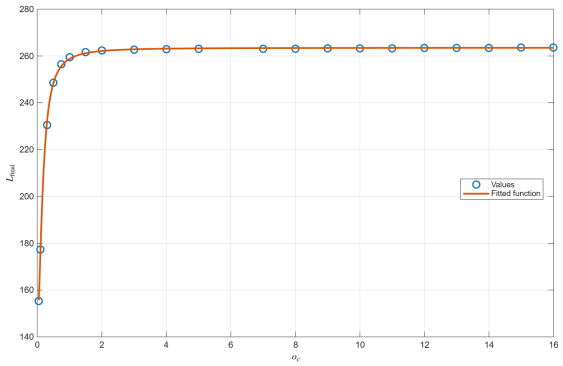
(b) Final domain length as a function of the oxygen concentration imposed at the boundaries

Figure 3.5: Influence of the boundary oxygen concentration on domain length

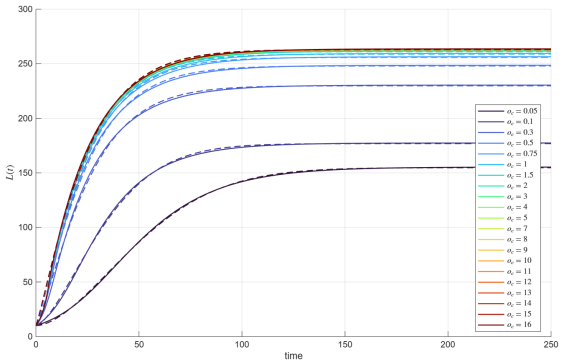
Figure 3.5a shows the domain length as a function of time for several values of o_c. For all the used values of o_c, we note that the domain reaches a plateau. The higher o_c is, the faster the plateau is reached and the higher it is. Figure 3.5b shows the value of this plateau as a function of o_c. In this figure, we note that the final length itself reaches a plateau: beyond a sufficient threshold, increasing o_c has no impact on the final length.

Looking at the equations of the system, we see that the only term which depends on o_c is $H(o)$ and o with the boundary conditions. As $\lim_{o_c \to +\infty} H(o_c) = 1$, when o_c becomes sufficiently high, cell equation depends on o_c only by the chemotactic term.

To characterize the dependence of the final length on o_c, the results are fitted by a function of the form $L_{final}(o_c) = B + \frac{A}{1+\left(\frac{o_c}{C}\right)^p}$, as shown in fig. 3.6a.



(a) Fit of the final domain length as a function of the boundary oxygen concentration



(b) Fit of the domain length as a function of the time

Figure 3.6: Fit of the domain length as a function of the boundary oxygen concentration and time

The fitted parameters are: $A = -119.449$, $B = 263.536$, $C = 0.172$, $p = 1.806$.

With this function, the coefficient of determination obtained is $R^2 = 0.99980474$, which indicates that the function provides a very good fit to the values.

We also fit the results of the temporal evolution of the length with a function of the form $L(t, o_c) = L_0 + (L_{final}(o_c) - L_0)\left[1 - e^{-(\frac{t}{\tau(o_c)})^{p(o_c)}}\right]$ in fig. 3.6b.

For τ and p, which are functions of o_c, we obtain:

$$\tau(o_c) = 25.200 + 87.316e^{-18.999o_c},$$
$$p(o_c) = 1.157 + 0.825e^{-8.079o_c}.$$

In this chapter, I introduced a reaction-cross-diffusion model on a growing domain. I performed a linear stability analysis of the non-autonomous system and extended the existing instability criterion to a two-species system with a non-diagonal diffusion matrix. I then extended the analysis to a three-species system. Finally, I introduced oxygen supply at the boundaries of the domain as a mechanism to limit tissue growth. The numerical results showed that oxygen limitation can lead to a saturation of the tissue length.

However, oxygen limitation is not the only mechanism that can lead to a finite final domain length. In the next chapter, I will introduce mechanical effects into the reaction-cross-diffusion model and investigate how mechanical constraints can affect tissue growth and pattern formation.

Chapter 4

Mechanical regulation of tissue growth

The second mechanism that will be investigated is mechanical regulation of tissue growth. So far, growth has changed the geometry of the domain, but we have not explicitly considered the mechanical consequences of this deformation. In a real tissue, growth induces deformation which, together with interactions with the surrounding environment, can generate mechanical stress.

Therefore, we introduce a mechanical feedback: instead of prescribing growth independently of mechanics, the local growth rate is assumed to depend on the stress.

4.1 Equations of the mechanochemical model

As in the previous chapter, we denote by $X \in [0, L_0]$ the material coordinate in the reference configuration. The current position of the material point initially located at X is denoted by $x = x(X, t)$. We introduce the displacement field $\xi(X, t)$, which represents the displacement of this material point from its reference position. Therefore,

$$x(X, t) = X + \xi(X, t). \tag{4.1}$$

The total deformation gradient is then defined as

$$F(X, t) = \frac{\partial x}{\partial X} = 1 + \frac{\partial \xi}{\partial X}. \tag{4.2}$$

F describes the actual local deformation of the tissue. If F is greater than 1, the tissue is locally extended, while if F is smaller than 1, it is locally contracted.

4.1.1 Cells dynamics

We consider a two-species reaction-cross-diffusion system on a growing domain:

$$\frac{\partial u}{\partial t} = \underbrace{\frac{1}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{d_1(u,v)}{F(X,t)}\frac{\partial u}{\partial X} + \frac{d_2(u,v)}{F(X,t)}\frac{\partial v}{\partial X}\right)}_{\text{Cross-diffusion}} - \underbrace{u\frac{1}{F(X,t)}\frac{\partial F}{\partial t}(X,t)}_{\text{Dilution}} + \underbrace{f(u,v)}_{\text{Reaction}}, \tag{4.3}$$

$$\frac{\partial v}{\partial t} = \underbrace{\frac{1}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{d_3(u,v)}{F(X,t)}\frac{\partial u}{\partial X} + \frac{d_4(u,v)}{F(X,t)}\frac{\partial v}{\partial X}\right)}_{\text{Cross-diffusion}} - \underbrace{v\frac{1}{F(X,t)}\frac{\partial F}{\partial t}(X,t)}_{\text{Dilution}} + \underbrace{g(u,v)}_{\text{Reaction}}. \tag{4.4}$$

These equations remain very similar to the ones that we had in the previous chapter, the only difference being that the geometry is no longer described directly by the growth variable μ, but by the total deformation gradient F. Hence, growth and deformation are no longer the same thing. The tissue may locally grow, but its actual deformation also depends on the mechanical forces and constraints acting on it. So F is not prescribed independently: it is determined by the mechanical part of the model, introduced in the next section.

4.1.2 Mechanical equations

We use the morphoelasticity assumption developed by Goriely [6], which is based on a multiplicative decomposition of the total deformation gradient. Thus, the total deformation gradient can be decomposed into the product of an elastic deformation gradient and a growth deformation gradient. The growth deformation gradient corresponds to the deformation that would be produced by growth alone and would occur in the absence of mechanical stresses. The elastic deformation gradient corrects the incompatibilities that may arise from growth alone and restores mechanical equilibrium. The multiplicative decomposition of the deformation gradient is illustrated in fig. 4.1.

$$\underbrace{F(X,t)}_{\substack{\text{Total}\\ \text{deformation}\\ \text{gradient}}} = \underbrace{A(X,t)}_{\substack{\text{Elastic}\\ \text{deformation}\\ \text{gradient}}}\ \underbrace{G(X,t)}_{\substack{\text{Growth}\\ \text{deformation}\\ \text{gradient}}}. \tag{4.5}$$

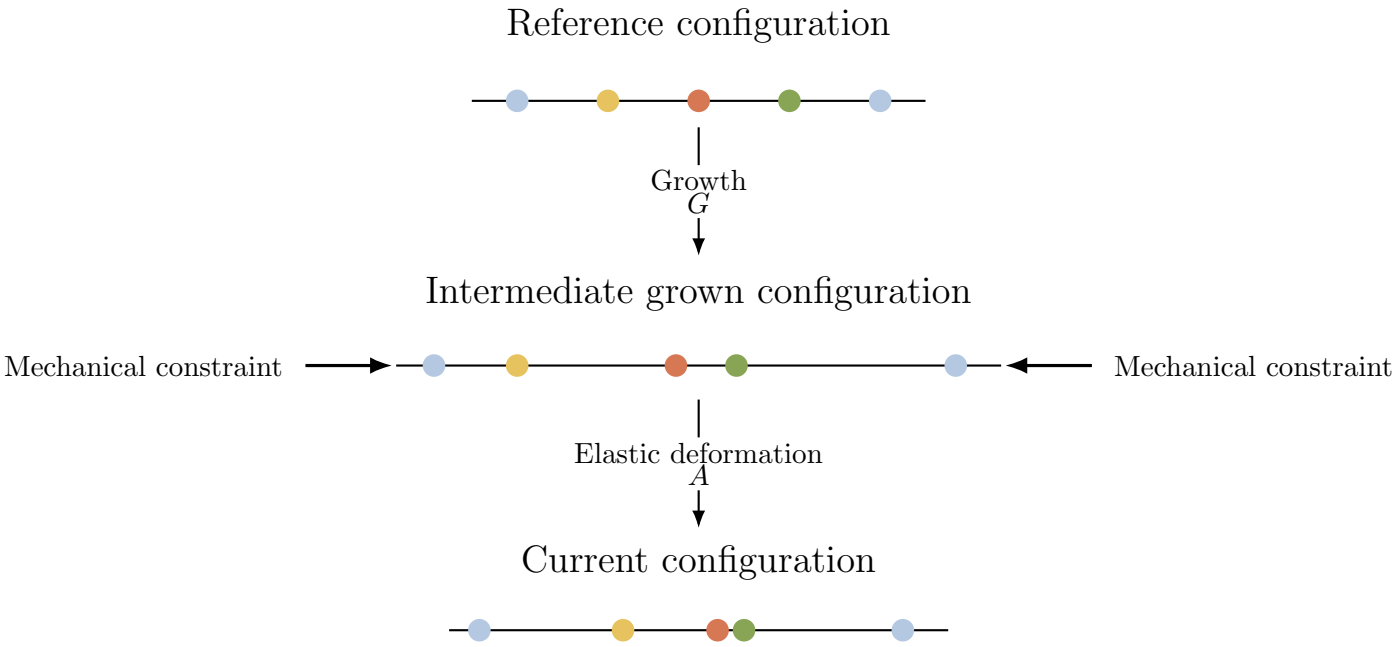


Figure 4.1: Schematic representation of the multiplicative decomposition of the deformation gradient. Growth maps the reference configuration onto the intermediate grown configuration through G, while the elastic deformation A maps the intermediate configuration onto the current configuration under mechanical constraints. The total deformation gradient is given by $F = AG$.

We then introduce a growth law which depends on the stress:

$$\frac{\partial G}{\partial t}(X,t) = S(t,u,v,\sigma)G(X,t). \tag{4.6}$$

Here, S represents the local tissue growth rate, whereas G represents the accumulated growth deformation. In this work, we assume that mechanical compression can inhibit tissue growth but does not directly induce tissue loss. The local tissue growth rate S is therefore assumed to remain non-negative.

As a first approximation, we assume that the tissue behaves as a linear elastic material. Growth deformation is assumed to be stress-free. Consequently, the mechanical stress σ depends only on the elastic deformation gradient A. Using Hooke's law, which is valid for small elastic deformations, we write

$$\sigma(X,t) = E(A(X,t) - 1), \tag{4.7}$$

with E Young's modulus. Therefore, $A = 1$ corresponds to a stress-free state, while $A > 1$ corresponds to tension and $A < 1$ to compression.

As the timescale of the mechanical relaxation is much shorter than the timescale of growth and cell proliferation, the material is assumed to be in quasi-static mechanical equilibrium.

$$\frac{\partial \sigma}{\partial X}(X,t) - \underbrace{(u(X,t)s_u + v(X,t)s_v)\xi(X,t)}_{\substack{\text{Body force} \\ \text{density}}} = 0. \tag{4.8}$$

The body force represents the mechanical interaction between the cells and their surrounding environment, such as the extracellular matrix. Each cell is subjected to a restoring force proportional to its displacement ξ, with s characterizing the strength of the mechanical coupling.

4.1.3 Boundary and initial conditions

We impose zero-flux boundary conditions on both sides:

$$\left(\frac{d_1(u,v)}{F(X,t)}\frac{\partial u}{\partial X} + \frac{d_2(u,v)}{F(X,t)}\frac{\partial v}{\partial X}\right)\Bigg|_{X=0,L_0} = 0, \tag{4.9}$$

$$\left(\frac{d_3(u,v)}{F(X,t)}\frac{\partial u}{\partial X} + \frac{d_4(u,v)}{F(X,t)}\frac{\partial v}{\partial X}\right)\Bigg|_{X=0,L_0} = 0. \tag{4.10}$$

We first consider free boundary conditions, corresponding to a tissue with no external traction applied at either end. The mechanical stress therefore vanishes at both boundaries:

$$\sigma(0,t) = \sigma(L_0,t) = 0. \tag{4.11}$$

Using equation (4.7), we obtain:

$$A(0,t) = A(L_0,t) = 1. \tag{4.12}$$

The initial conditions are prescribed as

$$u(X,0) = u_0(X), \quad v(X,0) = v_0(X). \tag{4.13}$$

We assume that initially, the tissue has not been displaced, and that no growth deformation has yet accumulated:

$$\xi(X,0) = 0, \quad G(X,0) = 1. \tag{4.14}$$

Consequently,

$$F(X,0) = 1. \tag{4.15}$$

We can note that, in the free-boundary case, mechanical stress arises from the non-zero body force density. Indeed, if $s_u = s_v = 0$, $\frac{\partial \sigma}{\partial X} = 0$. Combined with the boundary conditions $\sigma(0,t) = \sigma(L_0,t) = 0$, this implies that $\sigma(X,t) = 0$ throughout the domain. In this case, there is no mechanical feedback on growth, and the model reduces to the model without mechanics.

4.2 Numerical implementation

4.2.1 Differential algebraic equation

Since mechanical relaxation is assumed to be instantaneous compared with cell dynamics and tissue growth, the mechanical variables do not satisfy evolution equations. Instead, they must satisfy the mechanical equilibrium and compatibility constraints at each time. The coupled problem therefore takes the form of a differential-algebraic system.

Let $C_u = uF$ and $C_v = vF$. These variables allows the dilution term to be incorporated into the time derivative, which leads to a more convenient conservative formulation for the numerical implementation. Substituting these expressions into equation (4.3) and (4.4), we obtain:

$$\frac{\partial C_u}{\partial t}(X,t) = \frac{\partial}{\partial X}\left(\frac{d_1(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_u}{F}\right) + \frac{d_2(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_v}{F}\right)\right) + Ff(\frac{C_u}{F},\frac{C_v}{F}), \tag{4.16}$$

$$\frac{\partial C_v}{\partial t}(X,t) = \frac{\partial}{\partial X}\left(\frac{d_3(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_u}{F}\right) + \frac{d_4(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_v}{F}\right)\right) + Fg(\frac{C_u}{F},\frac{C_v}{F}). \tag{4.17}$$

The system becomes:

$$\begin{cases} C_{u,t} &= \frac{\partial}{\partial X}\left(\frac{d_1(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_u}{F}\right)+\frac{d_2(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_v}{F}\right)\right)+Ff(\frac{C_u}{F},\frac{C_v}{F}) \\ C_{v,t} &= \frac{\partial}{\partial X}\left(\frac{d_3(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_u}{F}\right)+\frac{d_4(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_v}{F}\right)\right)+Fg(\frac{C_u}{F},\frac{C_v}{F}) \\ G_t &= S(t,\frac{C_u}{F},\frac{C_v}{F},\sigma)G \\ 0 &= AG-1-\frac{\partial\xi}{\partial X} \\ 0 &= \left(\frac{C_u}{F}s_u+\frac{C_v}{F}s_v\right)\xi-\frac{\partial\sigma}{\partial X} \end{cases}. \tag{4.18}$$

The spatial domain is discretised using N nodes. The variables C_u, C_v, G and ξ are defined at the nodes, whereas the elastic deformation gradient A is defined at the $N-1$ interfaces. Now, the system can be written as:

$$\mathbf{M}\dot{\mathbf{y}} = \boldsymbol{\phi}(\mathbf{y},t), \tag{4.19}$$

with the mass matrix:

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_N & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I}_N & 0 & 0 & \\ 0 & 0 & \mathbf{I}_N & 0 & 0 \\ 0 & 0 & 0 & \mathbf{0}_{N-1} & 0 \\ 0 & 0 & 0 & 0 & \mathbf{0}_N \end{pmatrix} \tag{4.20}$$

$$\boldsymbol{\phi}(\mathbf{y},t) = \begin{pmatrix} \frac{\partial}{\partial X}\left(\frac{d_1(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_u}{F}\right)+\frac{d_2(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_v}{F}\right)\right)+Ff(\frac{C_u}{F},\frac{C_v}{F}) \\ \frac{\partial}{\partial X}\left(\frac{d_3(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_u}{F}\right)+\frac{d_4(\frac{C_u}{F},\frac{C_v}{F})}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{C_v}{F}\right)\right)+Fg(\frac{C_u}{F},\frac{C_v}{F}) \\ S(t,\frac{C_u}{F},\frac{C_v}{F},\sigma)G \\ AG-1-\frac{\partial\xi}{\partial X} \\ \left(\frac{C_u}{F}s_u+\frac{C_v}{F}s_v\right)\xi-\frac{\partial\sigma}{\partial X} \end{pmatrix} \tag{4.21}$$

and:

$$\mathbf{y} = \begin{pmatrix} C_u \\ C_v \\ G \\ A \\ \xi \end{pmatrix}. \tag{4.22}$$

C_u, C_v and G satisfy differential equations, whereas A and ξ are determined by algebraic constraints.

The formulation of the model for n cell populations is provided in Appendix D.

4.2.2 Algorithm

The numerical procedure used to solve the model is the following :

- **Step 1:** Definition of the parameters of the model, the initial conditions for C_u, C_v, G, $A_{i+\frac{1}{2}}$ and ξ, and the mass matrix $\mathbf{M}$.
- **Step 2:** Spatial discretization of the diffusion and mechanical terms in the right-hand side of the DAE system.
- **Step 3:** For each evaluation of the DAE right-hand side, reconstruction of the deformation gradient at the interfaces:

$$F_{i+\frac{1}{2}} = A_{i+\frac{1}{2}} G_{i+\frac{1}{2}}, \qquad G_{i+\frac{1}{2}} = \frac{G_i + G_{i+1}}{2}. \tag{4.23}$$

- **Step 4:** Interpolation of F from the interfaces to the nodes.
- **Step 5:** Calculation of the cell densities using $u = \frac{C_u}{F}$ and $v = \frac{C_v}{F}$.
- **Step 6:** Calculation of the mechanical stress at the interfaces using

$$\sigma_{i+\frac{1}{2}} = E\left(A_{i+\frac{1}{2}} - 1\right). \tag{4.24}$$

- **Step 7:** Evaluation of the reaction, diffusion, growth and mechanical terms, and assembly of the right-hand side $\boldsymbol{\phi}(\mathbf{y}, t)$.
- **Step 8:** Solution of the DAE system using ode15s.
- **Step 9:** Reconstruction of the physical position x using $x(X,t) = X + \xi(X,t)$.

4.3 Results for free-free boundary conditions

4.3.1 Stress-dependent growth law

We first define the dependence of the local tissue growth rate on the mechanical stress. We assume that compression inhibits tissue growth, but does not directly induce tissue loss. We therefore consider the following stress-dependent growth law:

$$S(\sigma) = \begin{cases} S_0, & \sigma \geq 0, \\ S_0\left(1 - \frac{|\sigma|}{\sigma_c}\right), & -\sigma_c < \sigma < 0, \\ 0, & \sigma \leq -\sigma_c, \end{cases} \tag{4.25}$$

where $S_0 > 0$ is the growth rate in the absence of mechanical inhibition and $\sigma_c > 0$ is the critical compressive stress above which tissue growth is completely inhibited.

When $\sigma \geq 0$, the tissue grows at rate S_0. For compressive stresses, the growth rate progressively decreases as the magnitude of the compression increases. When $\sigma = -\sigma_c$, growth is completely inhibited. For larger compressive stresses, the growth rate remains equal to zero.

4.3.2 Simulations

For the numerical simulations, we consider the Schnakenberg reaction kinetics introduced in section 2.3.1:

$$f(u,v) = a - u + u^2 v, \qquad g(u,v) = b - u^2 v. \tag{4.26}$$

The cross-diffusion matrix is chosen as

$$\mathbf{D}(u,v) = \begin{pmatrix} D_u & -\chi_0 \frac{ku}{k+v} \\ 0 & D_v \end{pmatrix}. \tag{4.27}$$

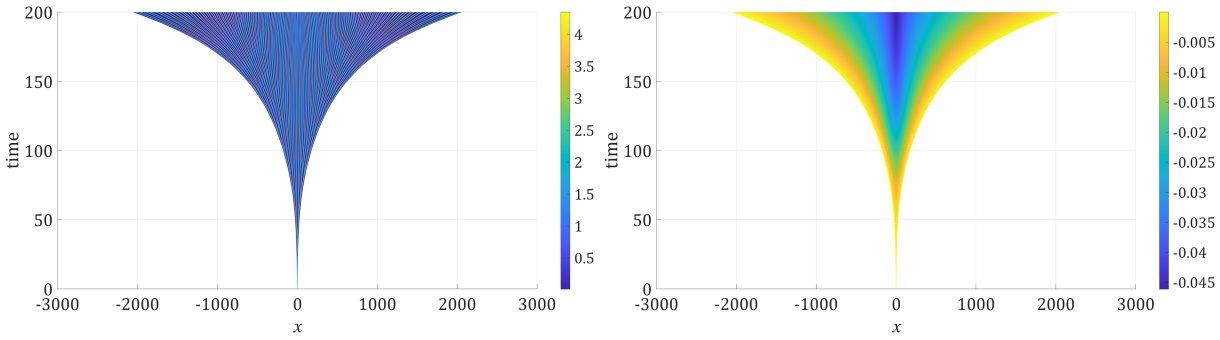
Results are shown in fig. 4.2. In fig. 4.2a, we see that as in the model without mechanics, peak-splitting events are observed, allowing new peaks to appear as the domain expands. However, contrary to the model without mechanics, these events no longer occur simultaneously throughout the domain. Peak-merging events are also observed, which were not present in the simulations without mechanical feedback.

This difference can be related to the coupling between growth and mechanics. Indeed, the actual deformation of the tissue is now determined by $F = AG$, and the mechanical stress also locally modifies the growth rate through S. Consequently, the expansion of the tissue is no longer determined by growth alone: regions subjected to different levels of compression may deform and grow at different rates. The different parts of the pattern therefore do not evolve under the same local conditions, which can explain why peak-splitting events occur at different times. The resulting local deformation can also modify the distance between neighbouring peaks and may lead to their merging. Therefore, the introduction of mechanics provides an additional mechanism through which the spatial pattern can reorganise during tissue growth.

Despite the mechanical feedback, and contrary to the oxygen model, the final length does not reach a finite value. Indeed, the local tissue growth rate is always greater than or equal to zero. Hence, in highly compressed regions, S is equal to zero so the tissue locally stops growing. But at the boundary, the chosen conditions impose $\sigma(0,t) = \sigma(L_0,t) = 0$, which can be seen in fig. 4.2b. This implies that at the

boundaries, S is always positive. Consequently, in the simulations considered here, the mechanically inhibited regions are not sufficient to compensate for the growth occurring in the less compressed regions, and the tissue continues to expand.

Thereby, with free boundaries, internal mechanical stress can modify the dynamics and the pattern, but in this parameter regime it does not provide a mechanism for obtaining a finite tissue size. This motivates introducing an external mechanical constraint on the tissue.



(a) Evolution of the cell population u (b) Evolution of the mechanical stress

Figure 4.2: Mechanochemical dynamics under free-free boundary conditions with parameters $a = 0.10$, $b = 1.1$, $D_u = 1$, $D_v = 40$, $\chi_0 = 0.01$, $k = 1$, $L_0 = 5$, $s_u = 10^{-4}$, $s_v = 10^{-5}$, $S_0 = 0.049082$ and $\sigma_c = 0.04$.

4.3.3 Extension of the constitutive law

So far, the tissue has been assumed to behave as an instantaneous linear elastic material, with $\sigma = E(A-1)$. However, biological tissues may exhibit time-dependent mechanical behaviour. To investigate whether the choice of constitutive law affects the mechanochemical dynamics, we extend the elastic relation to a general differential constitutive law of the form

$$a_2\frac{\partial^2\sigma}{\partial t^2} + a_1\frac{\partial\sigma}{\partial t} + a_0\sigma = b_2\frac{\partial^2\epsilon}{\partial t^2} + b_1\frac{\partial\epsilon}{\partial t} + b_0\epsilon, \tag{4.28}$$

where

$$\epsilon = \mathcal{E}(c_1, \ldots, c_n, F, G, \xi). \tag{4.29}$$

This formulation includes different mechanical behaviours depending on the values of the coefficients. In particular, the elastic model considered previously corresponds to an order-zero constitutive law, whereas first-order laws can be used to introduce viscoelastic effects.

The corresponding DAE formulations for the different orders are detailed in Appendix E.

4.4 Extension of the boundary conditions: rigid fixed walls

4.4.1 Model

We extend the general model of section 4.1 by introducing two rigid and fixed walls, one on each side of the domain. Initially, the domain has free-free boundary conditions, and as the tissue grows, it may come into contact with one of the walls. This contact generates a compressive reaction pressure, denoted by p_R on the right wall and p_L on the left wall. The coordinates of the walls are denoted by x_L^W and x_R^W. We denote by x_R^T and x_L^T the extremities of the tissue.

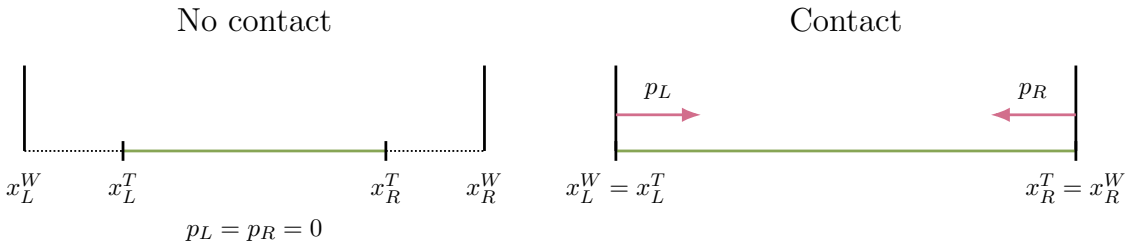


Figure 4.3: Schematic representation of the rigid-wall boundary conditions.

The tissue cannot penetrate the walls, so we impose the conditions

$$x_R^T \leq x_R^W \quad \text{and} \quad x_L^T \geq x_L^W. \tag{4.30}$$

When there is no contact,

$$x_R^W - x_R^T > 0, \quad x_L^T - x_L^W > 0 \quad \text{and} \quad p_R = p_L = 0. \tag{4.31}$$

At contact,

$$x_R^W - x_R^T = 0 \quad \text{and} \quad p_R \neq 0, \quad \text{or} \quad x_L^T - x_L^W = 0 \quad \text{and} \quad p_L \neq 0. \tag{4.32}$$

Therefore, in both cases, we have

$$(x_R^W - x_R^T)p_R = 0 \quad \text{and} \quad (x_L^T - x_L^W)p_L = 0. \tag{4.33}$$

with

$$x_R^W - x_R^T \geq 0, \qquad x_L^T - x_L^W \geq 0, \qquad p_R, p_L \geq 0. \tag{4.34}$$

Thus, the system still follows equations of section 4.1.1 and section 4.1.2, with at the boundaries:

$$(x_R^W - x_R^T(t))p_R(t) = 0, \tag{4.35}$$

$$(x_L^W - x_L^T(t))p_L(t) = 0, \tag{4.36}$$

$$\sigma(0,t) = -p_L(t), \tag{4.37}$$

$$\sigma(L_0,t) = -p_R(t). \tag{4.38}$$

4.4.2 Numerical method

The rigid walls are implemented using a penalty method. Initially, the tissue is located between the two walls and does not interact with them. Therefore, as long as the tissue boundaries do not reach the walls, the mechanical boundary conditions are equivalent to free boundary conditions.

At each time, the physical positions of the left and right boundaries of the tissue are computed as

$$x_L(t) = X_1 + \xi_1(t), \qquad x_R(t) = X_N + \xi_N(t). \tag{4.39}$$

The signed penetrations are defined by

$$z_L = x_L^W - x_L^T, \qquad z_R = x_R^T - x_R^W. \tag{4.40}$$

Thus, $z_L \leq 0$ and $z_R \leq 0$ correspond to the absence of penetration, whereas positive values indicate penetration into a wall.

To prevent this penetration, a penalty pressure is introduced at each boundary. Ideally, this pressure would be written as

$$p(z) = k_W[z]_+ = k_W \max(0,z), \tag{4.41}$$

where k_W is the penalty stiffness. In this way, the pressure is zero before contact and increases when the tissue attempts to penetrate the wall. The mechanical boundary conditions therefore become

$$\sigma(0,t) + p(z_L) = 0, \qquad \sigma(L_0,t) + p(z_R) = 0. \tag{4.42}$$

In the numerical implementation, the positive-part function is locally smoothed around $z = 0$ in order to avoid the non-differentiability of $\max(0, z)$ at first contact. To obtain a smoother numerical transition, we use the locally regularised positive-part function

$$[z]_+^\epsilon = \frac{1}{2}\left(z + \sqrt{z^2 + \phi(z)\epsilon^2}\right), \tag{4.43}$$

where

$$\phi(z) = \begin{cases} \delta\left(1 - \dfrac{z^2}{c_0^2}\right)^2, & |z| < c_0, \\ 0, & |z| \geq c_0. \end{cases} \tag{4.44}$$

Here, $\epsilon > 0$ controls the magnitude of the regularisation, $c_0 > 0$ defines the width of the transition region around first contact, and $\delta > 0$ controls the smoothing function within this region.

The penalty pressure is then defined as

$$p(z) = k_W [z]_+^\epsilon. \tag{4.45}$$

Outside the transition region $|z| < c_0$, this regularisation is exactly equal to $\max(0, z)$. Away from the transition region, the contact pressure is zero before contact and increases when the tissue penetrates the wall, while the transition around first contact is smoothed numerically.

An active-set method was also tested to handle the contact conditions with the rigid walls. This method consists in determining, at each time, which contact constraints are active, that is, which tissue boundaries are in contact with the walls. The mechanical boundary conditions are then adapted accordingly: a free boundary condition is used in the absence of contact, while the contact constraint is imposed when the wall is reached. The results obtained were similar to those obtained with the penalty method. In the following, only the results obtained with the penalty method are presented.

4.4.3 Simulations

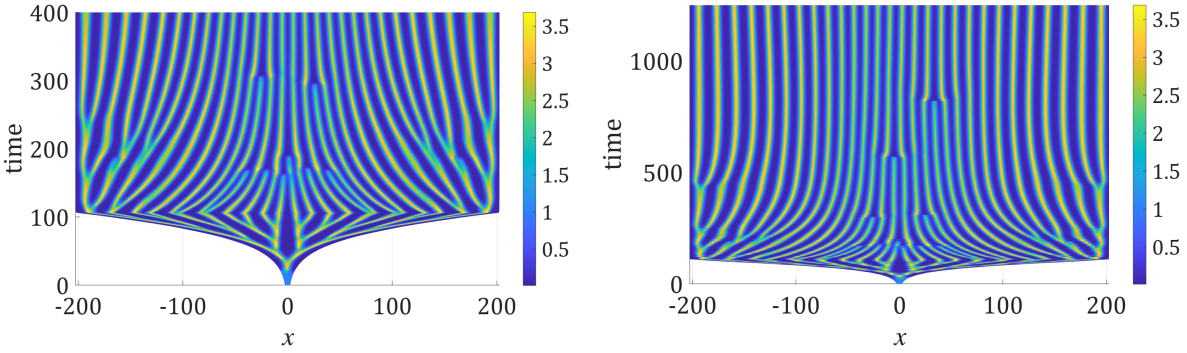
For the simulations, we use the same model as in section 4.3.2, i.e. a Schnakenberg model with a chemotactic term.

As a reminder, the growth law is given by

$$S(\sigma) = \begin{cases} S_0, & \sigma \geq 0, \\ S_0\left(1 - \frac{|\sigma|}{\sigma_c}\right), & -\sigma_c < \sigma < 0, \\ 0, & \sigma \leq -\sigma_c, \end{cases} \tag{4.46}$$

The numerical results obtained in the presence of two rigid walls are shown in fig. 4.4. The walls are located at $x = -200$ and $x = 205$. At the beginning of the simulation, the tissue is much smaller than the distance between the walls. The mechanical boundary conditions are therefore effectively free, and the tissue grows without interacting with the walls. During this first phase, until approximately $t = 110$, the domain expands rapidly and the patterns develop as in the free-free case of section 4.3.2. Then, when the tissue reaches the walls, its length stops increasing, while the pattern continues to evolve. Indeed, the pattern reorganises: near the boundaries, peaks continue to split, whereas near the centre, some peaks merge. These rearrangements become progressively less frequent, suggesting that the system is relaxing towards a new spatial organisation compatible with the constrained domain. This second phase lasts for a relatively long time, before a third phase is reached, during which the pattern, composed of vertical bands, appears to reach a stationary state.

Thus, these simulations show that reaching a finite domain length does not instantaneously lead to a stationary pattern: a transient reorganisation of the existing peaks is required before the final spatial organisation is reached.



(a) Evolution of the cell population u until $t_{\text{final}} = 400$

(b) Evolution of the cell population u until $t_{\text{final}} = 1500$

Figure 4.4: Mechanochemical dynamics in the presence of two rigid walls, with parameters $a = 0.10$, $b = 1.1$, $D_u = 1$, $D_v = 40$, $\chi_0 = 0.01$, $k = 1$, $L_0 = 5$, $s_u = 10^{-4}$, $s_v = 10^{-5}$, $S_0 = 0.049082$ and $\sigma_c = 0.04$. The rigid walls are located at $x = -200$ and $x = 205$.

In this chapter, I developed a reaction-cross-diffusion model that takes into account the mechanics of the tissue. The objective was to investigate whether mechanical effects could provide a mechanism to prevent the tissue from growing indefinitely. Two different boundary conditions were explored.

First, with free-free boundary conditions, the body forces generate internal mechanical stresses, which affect the local growth rate and lead to a reorganisation of the pattern. However, with the chosen growth law, the tissue cannot locally shrink, since the growth rate remains non-negative. Moreover, the free boundary conditions impose zero stress at both ends of the tissue, so that growth is never mechanically inhibited at the boundaries. As a consequence, in the simulations considered here, mechanical feedback modifies the pattern dynamics but is not sufficient to produce a finite tissue length.

In a second part, I introduced walls on both sides of the domain. When the tissue reaches the walls, this configuration leads to a finite domain length, and a slow reorganisation of the pattern occurs, before reaching what seems to be a final stationary pattern.

Chapter 5

Conclusion

The aim of this work was to study pattern formation in a growing tissue, while also considering the question of how this growth can be regulated. The objective was therefore, on the one hand, to develop tools for studying the stability of reaction-cross-diffusion systems on growing domains and, on the other hand, to explore different mechanisms that may limit tissue size.

First, the study of pattern formation on a fixed domain allowed us to recall the theoretical framework of Turing instability and to introduce the tools required for linear stability analysis. However, introducing domain growth modifies this analysis, since the system obtained after linearisation becomes non-autonomous. We therefore relied on the approach developed by Van Gorder for a two-species system with a diagonal diffusion matrix. This approach was first extended to the case of a non-diagonal diffusion matrix, thus allowing cross-diffusion to be taken into account. This extension provides both a sufficient condition for the growth of perturbations and a complementary condition guaranteeing their decay. A second extension was then developed for a three-species system, for which sufficient conditions guaranteeing the growth of spatial perturbations were obtained.

The second part of this work focused on limiting tissue growth. A first mechanism considered is based on oxygen availability. A model in which oxygen is supplied at the boundaries of the domain and consumed by the cells was constructed, in order to couple cell proliferation to the amount of oxygen available. The simulations show that this mechanism can lead to the saturation of tissue growth towards a final length. This final length increases with the oxygen concentration imposed at the boundaries, but itself tends towards a limiting value as this concentration increases.

Finally, a second regulation mechanism, of mechanical origin, was studied. To do so, tissue mechanics was introduced by distinguishing growth from elastic deformation, and the growth law was then coupled to the stress state of the tissue. The

results show that, in the case of free-free boundary conditions, the coupling between growth and stress considered here is not sufficient to lead to a finite final length. The interaction between the tissue and its environment therefore plays an important role. Introducing rigid walls constrains its expansion and also leads to a progressive reorganisation of the pattern when the tissue comes into contact with the walls.

Thus, this work highlights two complementary aspects of the problem: domain growth modifies the tools required to study pattern formation, while the dynamics of this growth itself depends on the biological and mechanical mechanisms taken into account.

The results obtained in this work nevertheless present several limitations. First, all the models studied are based on a one-dimensional representation of the tissue. This choice simplifies both the theoretical analysis and the numerical simulations, but does not allow the full geometrical complexity of biological tissues to be taken into account. This limitation is particularly important for the mechanical part, since the deformations and stresses that can arise in a two- or three-dimensional tissue are necessarily richer than in the one-dimensional case considered here.

The theoretical analysis developed on a growing domain also presents some limitations. In particular, in the case of a three-species system, the conditions obtained are only sufficient conditions guaranteeing the growth of certain spatial perturbations. They remain relatively restrictive, and their non-satisfaction does not imply the absence of instability. This limitation can notably be observed in the numerical simulations, in which patterns can form even when the theoretical conditions established do not predict the growth of the corresponding perturbations.

Finally, the mechanical model also remains deliberately simplified. In particular, it relies on a simple constitutive law and idealised boundary conditions. Another limitation concerns the numerical resolution of the model. The remeshing used in the previous models could not be retained when mechanics was introduced, since updating the mesh would have to satisfy simultaneously the constraints associated with remeshing and the mechanical equations. The simulations are therefore performed without remeshing. When the domain undergoes large deformations, the mesh can consequently become strongly non-uniform, which reduces the spatial resolution in some regions of the domain and limits the reliability of the simulations for large deformations.

More generally, the feedback between mechanical stress and growth considered in this work is not sufficient on its own to guarantee a saturation of tissue size. The re-

sults obtained notably highlight the importance of interactions with the environment in regulating this growth.

Several perspectives can be considered following this work. First, the theoretical analysis developed on a growing domain could be further investigated. In particular, it would be interesting to look for less restrictive instability criteria in the three-species case, capable of detecting a wider range of situations in which spatial perturbations can grow. More generally, the relationship between the predictions of the linear stability analysis and the patterns actually observed in nonlinear simulations could be studied in more detail. In addition, the criteria developed in this work cannot be directly applied to the oxygen and mechanical models introduced later. The oxygen model involves different boundary conditions for the different species, while the introduction of mechanics leads to a coupled system for which the homogeneous base state considered in the theoretical analysis is no longer available under the same conditions. An important perspective would therefore be to extend the stability analysis tools to these two models.

The two growth regulation mechanisms studied in this work were considered separately. A natural next step would therefore be to combine them within a single model, in which growth would depend both on oxygen availability and on the mechanical state of the tissue. These two mechanisms may indeed interact: tissue growth and deformation modify the oxygen distribution, while oxygen availability in turn influences cell proliferation and therefore the generation of mechanical stresses. Such a model would make it possible to study jointly the biochemical and mechanical effects that may regulate tissue size.

The mechanical part could also be further developed. In particular, implementing a remeshing method compatible with the mechanical equations would make it possible to maintain a satisfactory spatial resolution when the tissue undergoes large deformations, and thus to study regimes that are currently difficult to simulate. It would also be interesting to consider more realistic interactions with the environment than the idealised boundary conditions studied here, in order to better understand the role of these interactions in growth regulation.

Finally, a natural extension of this work would be to consider higher-dimensional domains. Moving to two dimensions would not only provide a closer representation of the geometry of real biological tissues, but would also make it possible to study a wider variety of spatial patterns, such as spots or stripes. It would also allow more complex mechanical deformations and stresses to be taken into account, which cannot arise within the one-dimensional framework considered here.

This work therefore highlights the importance of studying pattern formation and tissue growth regulation together. The theoretical and numerical developments presented here open the way towards models that more fully integrate the interactions between pattern formation, growth, nutrient availability and mechanics.

Appendix A

Numerical implementation

The numerical simulations presented in this work are performed in MATLAB. This appendix describes the spatial discretisation, time integration and remeshing procedures used in the different models.

A.1 Spatial discretisation

A.1.1 Diffusion and cross-diffusion terms

We consider the spatial discretisation of the diffusion and cross-diffusion terms. In the Lagrangian coordinate X, the diffusion term associated with the first species is written as

$$\frac{1}{\mu}\frac{\partial}{\partial X}\left(\frac{d_1(u,v)}{\mu}\frac{\partial u}{\partial X}+\frac{d_2(u,v)}{\mu}\frac{\partial v}{\partial X}\right). \tag{A.1}$$

The reference domain is discretised using a uniform grid

$$X_i=(i-1)\Delta X, \qquad i=1,\dots,N, \tag{A.2}$$

where $\Delta X = L_0/(N-1)$. We denote by u_i, v_i and μ_i the values of u, v and μ at X_i.

The flux is first evaluated at the interfaces $X_{i+1/2}$. The values of the variables at an interface are approximated by

$$u_{i+\frac{1}{2}}=\frac{u_i+u_{i+1}}{2}, \qquad v_{i+\frac{1}{2}}=\frac{v_i+v_{i+1}}{2}, \qquad \mu_{i+\frac{1}{2}}=\frac{\mu_i+\mu_{i+1}}{2}, \tag{A.3}$$

while the spatial gradients are approximated by centred finite differences,

$$\left.\frac{\partial u}{\partial X}\right|_{i+\frac{1}{2}}\simeq\frac{u_{i+1}-u_i}{\Delta X}, \qquad \left.\frac{\partial v}{\partial X}\right|_{i+\frac{1}{2}}\simeq\frac{v_{i+1}-v_i}{\Delta X}. \tag{A.4}$$

The flux associated with the first species is therefore

$$q^u_{i+\frac{1}{2}} = \frac{1}{\mu_{i+\frac{1}{2}}}\left[d_1\left(u_{i+\frac{1}{2}}, v_{i+\frac{1}{2}}\right)\frac{u_{i+1}-u_i}{\Delta X} + d_2\left(u_{i+\frac{1}{2}}, v_{i+\frac{1}{2}}\right)\frac{v_{i+1}-v_i}{\Delta X}\right]. \tag{A.5}$$

For an interior grid point, its divergence is approximated by

$$\left[\frac{1}{\mu}\frac{\partial q^u}{\partial X}\right]_i \simeq \frac{1}{\mu_i}\frac{q^u_{i+\frac{1}{2}} - q^u_{i-\frac{1}{2}}}{\Delta X}, \qquad i = 2, \ldots, N-1. \tag{A.6}$$

For zero-flux boundary conditions, the flux through the boundaries is set to zero. Since the boundary control volumes have width $\Delta X/2$, this gives

$$\left[\frac{1}{\mu}\frac{\partial q}{\partial X}\right]_1 \simeq \frac{2q_{\frac{3}{2}}}{\mu_1 \Delta X}, \qquad \left[\frac{1}{\mu}\frac{\partial q}{\partial X}\right]_N \simeq -\frac{2q_{N-\frac{1}{2}}}{\mu_N \Delta X}. \tag{A.7}$$

A.1.2 Mechanical equations

For the mechanochemical model introduced in Chapter 4, the elastic deformation gradient A is discretised at the interfaces between neighbouring grid points, while G and ξ are defined at the nodes. We therefore denote by $A_{i+\frac{1}{2}}$ the elastic deformation gradient at the interface between X_i and X_{i+1}.

When interface values of nodal quantities are required, they are obtained by averaging the values at the neighbouring nodes. In particular,

$$G_{i+\frac{1}{2}} = \frac{G_i + G_{i+1}}{2}. \tag{A.8}$$

The total deformation gradient at the interfaces is then reconstructed from the multiplicative decomposition $F = AG$ as

$$F_{i+\frac{1}{2}} = A_{i+\frac{1}{2}} G_{i+\frac{1}{2}}. \tag{A.9}$$

The geometric compatibility equation

$$AG = 1 + \frac{\partial \xi}{\partial X} \tag{A.10}$$

is discretised at each interface as

$$A_{i+\frac{1}{2}} G_{i+\frac{1}{2}} - 1 - \frac{\xi_{i+1} - \xi_i}{\Delta X} = 0, \qquad i = 1, \ldots, N-1. \tag{A.11}$$

The strain and the stress are also evaluated at the interfaces:

$$\epsilon_{i+\frac{1}{2}} = A_{i+\frac{1}{2}} - 1, \qquad \sigma_{i+\frac{1}{2}} = E\epsilon_{i+\frac{1}{2}}. \tag{A.12}$$

The other constitutive laws considered in this work are treated in the same way, with the corresponding constitutive variables evaluated at the interfaces.

The mechanical equilibrium equation is discretised at the interior nodes. The spatial derivative of the stress is approximated from the two adjacent interface values:

$$\left.\frac{\partial \sigma}{\partial X}\right|_i \simeq \frac{\sigma_{i+\frac{1}{2}} - \sigma_{i-\frac{1}{2}}}{\Delta X}. \tag{A.13}$$

Therefore, for the mechanical equilibrium equation used in this work,

$$\frac{\partial \sigma}{\partial X} - (\mathbf{s}^T \mathbf{u})\xi = 0, \tag{A.14}$$

the discrete equation at an interior node is

$$\frac{\sigma_{i+\frac{1}{2}} - \sigma_{i-\frac{1}{2}}}{\Delta X} - (\mathbf{s}^T \mathbf{u}_i)\xi_i = 0, \qquad i = 2, \ldots, N-1. \tag{A.15}$$

The equations at the two boundary nodes are replaced by the corresponding mechanical boundary conditions. Free boundary conditions are implemented by imposing zero stress at the two external interfaces. The rigid-wall conditions are implemented through the penalty formulation described in the corresponding section.

A.2 Time integration

After spatial discretisation, the reaction-cross-diffusion models are reduced to systems of ordinary differential equations. These systems are integrated in time using the MATLAB solver `ode15s`. A sparsity pattern for the Jacobian is provided to the solver through the `JPattern` option, in order to reduce the computational cost.

For the simulations of the reaction-cross-diffusion system on a growing domain presented in Chapter 3, the relative and absolute tolerances are set to 10^{-6} and 10^{-8}, respectively. The simulations are generally performed using between 1000 and 2000 spatial grid points, depending on the model considered. In the oxygen model, 1000 grid points are used.

For the mechanochemical model introduced in Chapter 4, the spatial discretisation leads to a differential-algebraic system of the form

$$M\dot{\mathbf{y}} = \boldsymbol{\phi}(t, \mathbf{y}), \tag{A.16}$$

where the differential variables describe the evolution of the species concentrations and growth, while the mechanical variables satisfy algebraic constraints. The system is also solved using `ode15s`, with the mass matrix M supplied to the solver. The relative and absolute tolerances are set to 10^{-6} and 10^{-8}, respectively. The mechanical simulations use 8000 spatial grid points.

A.3 Remeshing

As the domain grows, the spatial grid can become increasingly stretched, which reduces the spatial resolution of the numerical solution. To avoid this problem, a remeshing procedure is applied at regular time intervals in the simulations without mechanics.

At each remeshing step, the current physical position is reconstructed from the local stretch μ according to

$$x(X,t) = \int_0^X \mu(\tilde{X},t)\,\mathrm{d}\tilde{X}. \tag{A.17}$$

The current domain length is therefore given by

$$L(t) = x(L_0,t). \tag{A.18}$$

A new uniform grid containing the same number of grid points is then introduced on the current physical domain $[0, L(t)]$. The species concentrations are linearly interpolated from the deformed grid onto this new uniform grid.

The new configuration is subsequently taken as the reference configuration for the next time interval. Consequently, the local stretch is reset to

$$\mu(X,t) = 1, \tag{A.19}$$

while the length of the reference domain is updated to the current physical length. The time integration is then restarted from the interpolated solution. For the oxygen model, the prescribed Dirichlet values of the oxygen concentration are imposed again at the two boundaries after interpolation.

This procedure maintains an approximately uniform spatial resolution as the domain grows. In the oxygen simulations presented in Chapter 3, remeshing is performed every 0.5 time units.

The remeshing procedure is not used for the mechanochemical model of Chapter 4. In this case, updating the reference configuration would also require the mechanical variables to satisfy the compatibility and equilibrium equations on the new mesh. The mechanical simulations are therefore performed without remeshing.

Appendix B

Verification of the instability criterion in the case of D diagonal

If $d_2^* = d_3^* = 0$, then:

$$\mathbf{B}_k(t) = \begin{pmatrix} -\frac{d_1^*(t)}{(\mu^*(t))^2}\rho_k + A_{11}(t) & A_{12}(t) \\ A_{21}(t) & -\frac{d_4^*(t)}{(\mu^*(t))^2}\rho_k + A_{22}(t) \end{pmatrix}. \tag{B.1}$$

Then:

$$\det(\mathbf{B}_k(t)) < b_{12}(t)\frac{d}{dt}\left(\frac{b_{11}(t)}{b_{12}(t)}\right) \tag{B.2}$$

$$\Leftrightarrow \left(-\frac{d_1^*(t)}{(\mu^*(t))^2}\rho_k + A_{11}(t)\right)\left(-\frac{d_4^*(t)}{(\mu^*(t))^2}\rho_k + A_{22}(t)\right) - A_{12}(t)A_{21}(t)$$
$$< A_{12}(t)\frac{d}{dt}\left(\frac{-\frac{d_1^*(t)}{(\mu^*(t))^2}\rho_k + A_{11}(t)}{A_{12}(t)}\right) \tag{B.3}$$

$$\Leftrightarrow \frac{d_1^*(t)d_4^*(t)}{(\mu^*(t))^4}\rho_k^2 - \frac{d_1^*(t)}{(\mu^*(t))^2}\rho_k A_{22}(t) - \frac{d_4^*(t)}{(\mu^*(t))^2}\rho_k A_{11}(t) + A_{11}(t)A_{22}(t) - A_{12}(t)A_{21}(t)$$
$$< A_{12}(t)\frac{d}{dt}\left(\frac{(\mu^*(t))^2 A_{11}(t) - d_1^*(t)\rho_k}{A_{12}(t)(\mu^*(t))^2}\right) \tag{B.4}$$

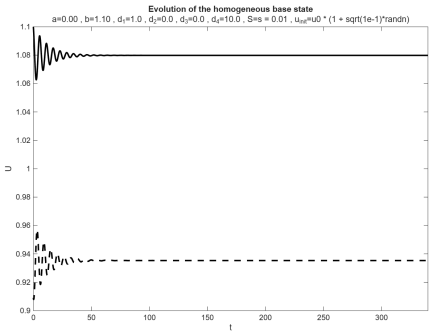
$$\Leftrightarrow \frac{d_1^*(t)d_4^*(t)}{(\mu^*(t))^4}\rho_k^2 - \frac{d_1^*(t)A_{22}(t) + d_4^*(t)A_{11}(t)}{(\mu^*(t))^2}\rho_k + \det(\mathbf{A}(t)) \tag{B.5}$$
$$< A_{12}(t)\frac{d}{dt}\left(\frac{(\mu^*(t))^2 A_{11}(t) - d_1^*(t)\rho_k}{A_{12}(t)(\mu^*(t))^2}\right).$$

This criterion is indeed the one found in Van Gorder's article [9].

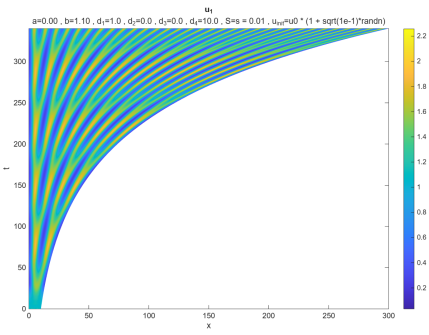
Appendix C

Numerical validation

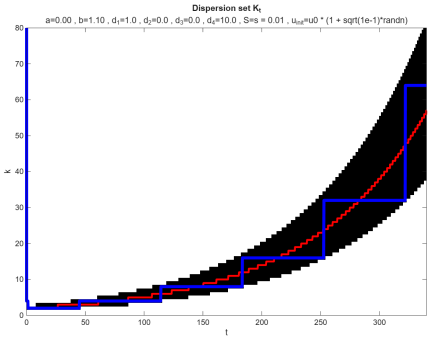
The aim of this part is to reproduce the numerical results obtained by Van Gorder et al. [9] with a Schnakenberg model in order to verify the numerical implementation of the instability criterion and the identification of the maximally growing mode.



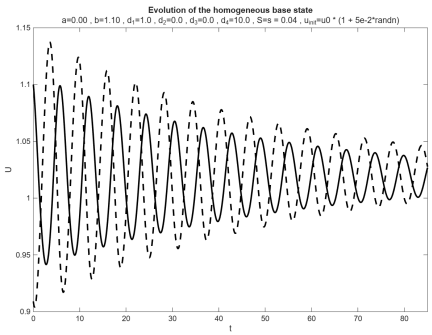
(a) i



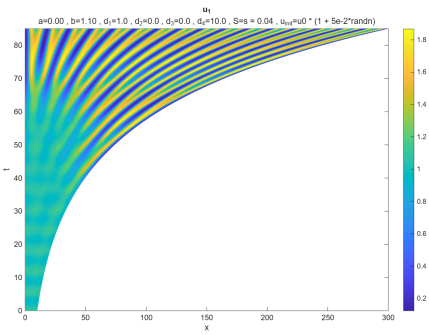
(a) ii



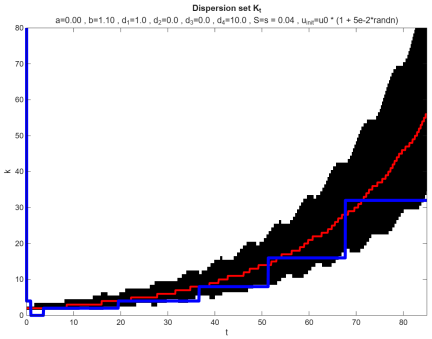
(a) iii



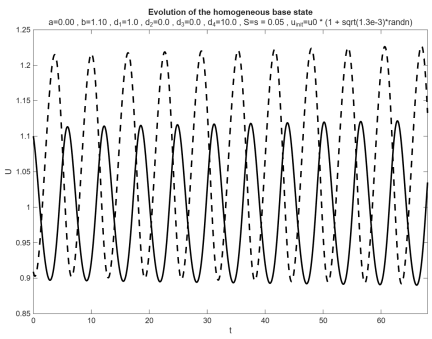
(b) i



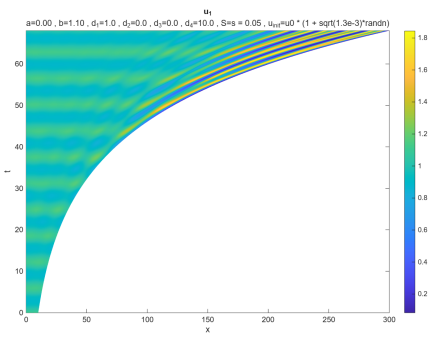
(b) ii



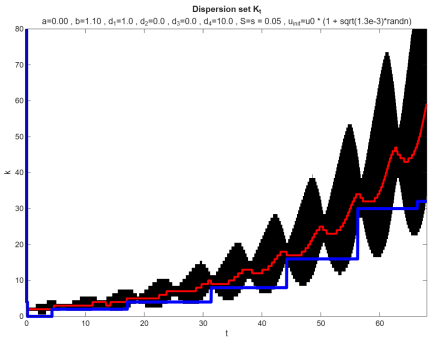
(b) iii



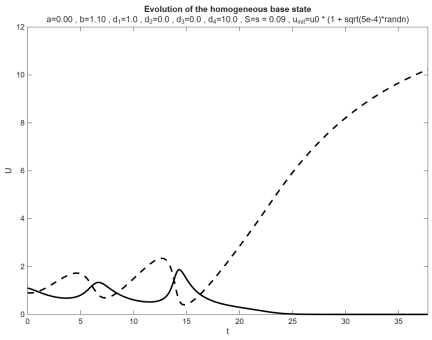
(c) i



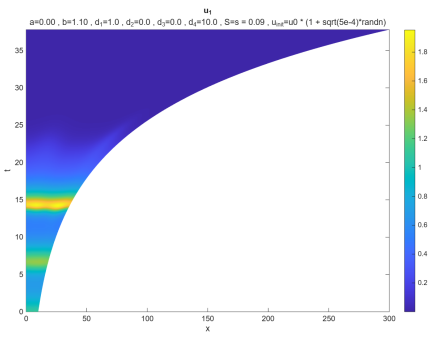
(c) ii



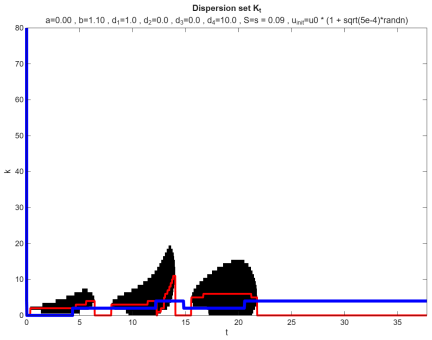
(c) iii



(d) i



(d) ii



(d) iii

Figure C.1: Simulations for 2 species with parameters $a = 0.1$, $b = 1.1$, $D_u = 1$, $D_v = 10$, and $L_0 = 10$.

Appendix D

Generalization of the mechanical model to n species

We then consider a reaction-cross-diffusion system of n species, on a growing one-dimensional domain:

$$\forall i \in [\![1,n]\!] \quad \frac{\partial c_i}{\partial t}(X,t) = \underbrace{f_i(c_1(X,t),...,c_n(X,t))}_{\text{Reaction}} - \underbrace{c_i(X,t)\frac{1}{F(X,t)}\frac{\partial F}{\partial t}(X,t)}_{\substack{\text{Dilution due to} \\ \text{tissue growth}}}$$

$$+ \underbrace{\frac{1}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{1}{F(X,t)}\sum_{j=1}^{n} D_{ij}(c_1(X,t),...,c_n(X,t))\frac{\partial c_j}{\partial X}(X,t)\right)}_{\text{Diffusion}}, \tag{D.1}$$

$$\frac{\partial G}{\partial t}(X,t) = S(t,c_1,...,c_n,\sigma)G(X,t). \tag{D.2}$$

The mechanical equations become:

$$\underbrace{F(X,t)}_{\substack{\text{Total} \\ \text{deformation}}} = 1 + \frac{\partial \xi}{\partial X}(X,t) = \underbrace{A(X,t)}_{\substack{\text{Elastic} \\ \text{deformation}}} \underbrace{G(X,t)}_{\substack{\text{Growth} \\ \text{deformation}}}, \tag{D.3}$$

$$\sigma(X,t) = E(A(X,t)-1), \tag{D.4}$$

$$\frac{\partial \sigma}{\partial X}(X,t) - \underbrace{\sum_{j=1}^{n} c_j(X,t)s_j\xi(X,t)}_{\substack{\text{Body force} \\ \text{density}}} = 0. \tag{D.5}$$

Let $C_i = c_i F$. The spatial domain is discretised using N nodes. The variables C_1, ..., C_n, G and ξ are defined at the nodes, whereas the elastic deformation gradient A is defined at the $N-1$ interfaces. Now, the DAE can be written as:

$$\mathbf{M}\dot{\mathbf{y}} = \boldsymbol{\phi}(\mathbf{y}, t) \tag{D.6}$$

with the mass matrix:

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_N & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I}_N & \ddots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & \mathbf{I}_N & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & \mathbf{I}_N & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \mathbf{0}_{N-1} & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & \mathbf{0}_N \end{pmatrix} \tag{D.7}$$

$$\boldsymbol{\phi}(\mathbf{y}, t) = \begin{pmatrix} F(X,t) f_1(\frac{C_1}{F}, ..., \frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)} \sum_{j=1}^{n} D_{1j}(\frac{C_1}{F}, ..., \frac{C_n}{F}) \frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ \vdots \\ F(X,t) f_n(\frac{C_1}{F}, ..., \frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)} \sum_{j=1}^{n} D_{nj}(\frac{C_1}{F}, ..., \frac{C_n}{F}) \frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ S(t, \frac{C_1}{F}, ..., \frac{C_n}{F}, \sigma) G \\ AG - 1 - \frac{\partial \xi}{\partial X} \\ \left(\sum_{j=1}^{n} \frac{C_j}{F} s_j\right) \xi - \frac{\partial \sigma}{\partial X} \end{pmatrix} \tag{D.8}$$

and:

$$\mathbf{y} = \begin{pmatrix} C_1 \\ \vdots \\ C_n \\ G \\ A \\ \xi \end{pmatrix}. \tag{D.9}$$

Appendix E

Constitutive laws

The aim of this appendix is to generalize the constitutive law introduced in Chapter 4. We consider the following system:

$$\frac{\partial c_i}{\partial t}(X,t) = \underbrace{f_i(c_1(X,t),...,c_n(X,t))}_{\text{Reaction}} - \underbrace{c_i(X,t)\frac{1}{F(X,t)}\frac{\partial F}{\partial t}(X,t)}_{\substack{\text{Dilution due to} \\ \text{tissue growth}}} + \underbrace{\frac{1}{F(X,t)}\frac{\partial}{\partial X}\left(\frac{1}{F(X,t)}\sum_{j=1}^{n} D_{ij}(c_1(X,t),...,c_n(X,t))\frac{\partial c_j}{\partial X}(X,t)\right)}_{\text{Diffusion}}, \quad \text{(E.1)}$$

$$\frac{\partial G}{\partial t}(X,t) = S(t,c_1,...,c_n,\sigma)G(X,t), \quad \text{(E.2)}$$

$$\underbrace{F(X,t)}_{\substack{\text{Total} \\ \text{deformation}}} = 1 + \frac{\partial \xi}{\partial X}(X,t) = \underbrace{A(X,t)}_{\substack{\text{Elastic} \\ \text{deformation}}}\ \underbrace{G(X,t)}_{\substack{\text{Growth} \\ \text{deformation}}}, \quad \text{(E.3)}$$

$$a_2\frac{\partial^2\sigma}{\partial t^2} + a_1\frac{\partial\sigma}{\partial t} + a_0\sigma(X,t) = b_2\frac{\partial^2\epsilon}{\partial t^2} + b_1\frac{\partial\epsilon}{\partial t} + b_0\epsilon(X,t), \quad \text{(E.4)}$$

$$\epsilon = \mathcal{E}\left(\frac{C_1}{F},...,\frac{C_n}{F},F,G,\xi\right), \quad \text{(E.5)}$$

$$\frac{\partial\sigma}{\partial X}(X,t) - \underbrace{\sum_{j=1}^{n} c_j(X,t)s_j\xi(X,t)}_{\substack{\text{Body force} \\ \text{density}}} = 0. \quad \text{(E.6)}$$

E.1 Order 0

We first consider the following law:

$$a_0\sigma(X,t) = b_0\epsilon(X,t) \tag{E.7}$$

with $a_0 \neq 0$.

Then,

$$\sigma(X,t) = \frac{b_0}{a_0}\epsilon(X,t). \tag{E.8}$$

Now, the DAE can be written as:

$$\mathbf{M}\dot{\mathbf{y}} = \boldsymbol{\phi}(\mathbf{y},t) \tag{E.9}$$

with the mass matrix:

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_N & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I}_N & \ddots & \vdots & \vdots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & \mathbf{I}_N & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & \mathbf{I}_N & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \mathbf{0}_{N-1} & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & \mathbf{0}_N \end{pmatrix} \tag{E.10}$$

$$\boldsymbol{\phi}(\mathbf{y},t) = \begin{pmatrix} F(X,t)f_1(\frac{C_1}{F},...,\frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)}\sum_{j=1}^{n} D_{1j}(\frac{C_1}{F},...,\frac{C_n}{F})\frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ \vdots \\ F(X,t)f_n(\frac{C_1}{F},...,\frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)}\sum_{j=1}^{n} D_{Nj}(\frac{C_1}{F},...,\frac{C_n}{F})\frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ S(t,\frac{C_1}{F},...,\frac{C_n}{F},\frac{b_0}{a_0}\epsilon)G \\ AG - 1 - \frac{\partial \xi}{\partial X} \\ \left(\sum_{j=1}^{n}\frac{C_j}{F}s_j\right)\xi - \frac{b_0}{a_0}\frac{\partial \epsilon}{\partial X} \end{pmatrix} \tag{E.11}$$

and:

$$\mathbf{y} = \begin{pmatrix} C_1 \\ \vdots \\ C_n \\ G \\ A \\ \xi \end{pmatrix}. \tag{E.12}$$

E.2 Order 1 with $a_1 \neq 0$

Then, we take the following law:

$$a_1 \frac{\partial \sigma}{\partial t} + a_0 \sigma = b_1 \frac{\partial \epsilon}{\partial t} + b_0 \epsilon \tag{E.13}$$

with $a_1 \neq 0$.

We introduce $q = a_1\sigma - b_1\epsilon$ as a new system variable. Then:

$$\frac{\partial q}{\partial t} = a_1 \frac{\partial \sigma}{\partial t} - b_1 \frac{\partial \epsilon}{\partial t} \tag{E.14}$$

and the mechanical stress law can be written as:

$$\frac{\partial q}{\partial t} = -a_0\sigma + b_0\epsilon = -\frac{a_0}{a_1}(q + b_1\epsilon) + b_0\epsilon = -\frac{a_0}{a_1}q + (b_0 - \frac{a_0}{a_1}b_1)\epsilon. \tag{E.15}$$

Now, the DAE can be written as:

$$\mathbf{M}\dot{\mathbf{y}} = \boldsymbol{\phi}(\mathbf{y}, t) \tag{E.16}$$

with the mass matrix:

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_N & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I}_N & \ddots & \vdots & \vdots & \vdots & 0 & 0 \\ \vdots & \ddots & \ddots & 0 & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & \mathbf{I}_N & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & \mathbf{I}_N & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \mathbf{0}_{N-1} & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & \mathbf{0}_N & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & \mathbf{I}_N \end{pmatrix} \tag{E.17}$$

$$\boldsymbol{\phi}(\mathbf{y}, t) = \begin{pmatrix} F(X,t) f_1(\frac{C_1}{F}, ..., \frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)} \sum_{j=1}^{n} D_{1j}(\frac{C_1}{F}, ..., \frac{C_n}{F}) \frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ \vdots \\ F(X,t) f_n(\frac{C_1}{F}, ..., \frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)} \sum_{j=1}^{n} D_{Nj}(\frac{C_1}{F}, ..., \frac{C_n}{F}) \frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ S(t, \frac{C_1}{F}, ..., \frac{C_n}{F}, \frac{1}{a_1}(q + b_1\epsilon))G \\ AG - 1 - \frac{\partial \xi}{\partial X} \\ \left(\sum_{j=1}^{n} \frac{C_j}{F} s_j\right)\xi - \frac{1}{a_1}\frac{\partial}{\partial X}\left[q + b_1\epsilon\right] \\ -\frac{a_0}{a_1}q + (b_0 - \frac{a_0}{a_1}b_1)\epsilon \end{pmatrix} \tag{E.18}$$

and:

$$\mathbf{y} = \begin{pmatrix} C_1 \\ \vdots \\ C_n \\ G \\ A \\ \xi \\ q \end{pmatrix}. \tag{E.19}$$

E.3 Order 1 with $a_1 = 0$

Then, we take the following law:

$$a_0\sigma = b_1\frac{\partial\epsilon}{\partial t} + b_0\epsilon. \tag{E.20}$$

Now, the DAE can be written as:

$$\mathbf{M}\dot{\mathbf{y}} = \boldsymbol{\phi}(\mathbf{y}, t) \tag{E.21}$$

with the mass matrix:

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_N & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I}_N & \ddots & \vdots & \vdots & \vdots & 0 & 0 & 0 \\ \vdots & \ddots & \ddots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & \mathbf{I}_N & 0 & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & \mathbf{I}_N & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \mathbf{0}_{N-1} & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & \mathbf{0}_N & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & b_1\mathbf{I}_N & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{0}_N \end{pmatrix} \tag{E.22}$$

$$\boldsymbol{\phi}(\mathbf{y},t) = \begin{pmatrix} F(X,t)f_1(\frac{C_1}{F},...,\frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)}\sum_{j=1}^{n} D_{1j}(\frac{C_1}{F},...,\frac{C_n}{F})\frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ \vdots \\ F(X,t)f_n(\frac{C_1}{F},...,\frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)}\sum_{j=1}^{n} D_{Nj}(\frac{C_1}{F},...,\frac{C_n}{F})\frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ S(t,\frac{C_1}{F},...,\frac{C_n}{F},\sigma)G \\ AG - 1 - \frac{\partial \xi}{\partial X} \\ \left(\sum_{j=1}^{n}\frac{C_j}{F}s_j\right)\xi - \frac{\partial \sigma}{\partial X} \\ a_0\sigma - b_0\epsilon \\ \epsilon - \mathcal{E}(\frac{C_1}{F},...,\frac{C_n}{F},F,G,\xi) \end{pmatrix} \tag{E.23}$$

and:

$$\mathbf{y} = \begin{pmatrix} C_1 \\ \vdots \\ C_n \\ G \\ A \\ \xi \\ \epsilon \\ \sigma \end{pmatrix}. \tag{E.24}$$

Remark: For the numerical implementation, $\mathcal{E}$ must depend explicitly on A.

E.4 Order 2

Now, we take the following law:

$$a_2\frac{\partial^2\sigma}{\partial t^2} + a_1\frac{\partial\sigma}{\partial t} + a_0\sigma = b_2\frac{\partial^2\epsilon}{\partial t^2} + b_1\frac{\partial\epsilon}{\partial t} + b_0\epsilon. \tag{E.25}$$

Let $r = \sigma_t$ and $z = \epsilon_t$. Then mechanical stress law can be written as:

$$\begin{cases} \sigma_t = r, \\ \epsilon_t = z, \\ a_2 r_t - b_2 z_t = -a_1 r - a_0\sigma + b_1 z + b_0\epsilon \end{cases} \tag{E.26}$$

Now, the DAE can be written as:

$$\mathbf{M}\dot{\mathbf{y}} = \boldsymbol{\phi}(\mathbf{y}, t) \tag{E.27}$$

with the mass matrix:

$$\mathbf{M} = \begin{pmatrix} \mathbf{I}_N & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I}_N & \ddots & \vdots & \vdots & \vdots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \ddots & \ddots & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & \mathbf{I}_N & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & \mathbf{I}_N & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & \mathbf{0}_{N-1} & 0 & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & \mathbf{0}_N & 0 & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & \mathbf{I}_N & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I}_N & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{0}_N & 0 \\ 0 & \cdots & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a_2\mathbf{I}_N & -b_2\mathbf{I}_N \end{pmatrix} \tag{E.28}$$

$$\boldsymbol{\phi}(\mathbf{y}, t) = \begin{pmatrix} F(X,t) f_1(\frac{C_1}{F}, ..., \frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)} \sum_{j=1}^{n} D_{1j}(\frac{C_1}{F}, ..., \frac{C_n}{F}) \frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ \vdots \\ F(X,t) f_n(\frac{C_1}{F}, ..., \frac{C_n}{F}) + \frac{\partial}{\partial X}\left[\frac{1}{F(X,t)} \sum_{j=1}^{n} D_{nj}(\frac{C_1}{F}, ..., \frac{C_n}{F}) \frac{\partial}{\partial X}\left(\frac{C_j}{F}\right)\right] \\ S(t, \frac{C_1}{F}, ..., \frac{C_n}{F}, \sigma) G \\ AG - 1 - \frac{\partial \xi}{\partial X} \\ \left(\sum_{j=1}^{n} \frac{C_j}{F} s_j\right) \xi - \frac{\partial \sigma}{\partial X} \\ r \\ z \\ \epsilon - \mathcal{E}(\frac{C_1}{F}, ..., \frac{C_n}{F}, F, G, \xi) \\ -a_1 r - a_0 \sigma + b_1 z + b_0 \epsilon \end{pmatrix} \tag{E.29}$$

and:

$$\mathbf{y} = \begin{pmatrix} C_1 \\ \vdots \\ C_n \\ G \\ A \\ \xi \\ \sigma \\ \epsilon \\ r \\ z \end{pmatrix}. \tag{E.30}$$

Remark: For the numerical implementation, $\mathcal{E}$ must depend on A.

Appendix F

Details of the calculations - Linearization and mode decomposition on a growing domain

F.1 Perturbation analysis

We perturb the system (3.7) - (3.9) by writing $u = u^*(t) + \epsilon U(X,t)$, $v = v^*(t) + \epsilon V(X,t)$ and $\mu = \mu^*(t) + \epsilon M(X,t)$ for $|\epsilon| \ll 1$.

Substituting these into (3.3), we obtain:

$$\begin{aligned}\frac{\partial u^*}{\partial t} + \epsilon\frac{\partial U(X,t)}{\partial t} = \frac{1}{\mu(X,t)}\frac{\partial}{\partial X}\Bigg[&\frac{d_1(u^*(t)+\epsilon U(X,t), v^*(t)+\epsilon V(X,t))}{\mu(X,t)}\epsilon\frac{\partial U(X,t)}{\partial X}\\ &+ \frac{d_2(u^*(t)+\epsilon U(X,t), v^*(t)+\epsilon V(X,t))}{\mu(X,t)}\epsilon\frac{\partial V(X,t)}{\partial X}\Bigg]\\ &- u^* S(t, u^*(t)+\epsilon U(X,t), v^*(t)+\epsilon V(X,t))\\ &- \epsilon U(X,t) S(t, u^*(t)+\epsilon U(X,t), v^*(t)+\epsilon V(X,t))\\ &+ f(u^*+\epsilon U(X,t), v^*+\epsilon V(X,t)).\end{aligned} \tag{F.1}$$

Now, expanding f, S and $\frac{1}{\mu}$ to the first order, we get:

$$f(u^*+\epsilon U(X,t), v^*+\epsilon V(X,t)) = f(u^*,v^*) + \epsilon\left(\frac{\partial f}{\partial u}(u^*,v^*)U + \frac{\partial f}{\partial v}(u^*,v^*)V\right) + O(\epsilon^2), \tag{F.2}$$

$$S(t,u^*+\epsilon U(X,t),v^*+\epsilon V(X,t)) = S(t,u^*,v^*) + \epsilon\left(\frac{\partial S}{\partial u}(t,u^*,v^*)U + \frac{\partial S}{\partial v}(t,u^*,v^*)V\right) + O(\epsilon^2), \tag{F.3}$$

$$\frac{1}{\mu(X,t)} = \frac{1}{\mu^*+\epsilon M} = \frac{1}{\mu^*} - \epsilon\frac{M}{(\mu^*)^2} + O(\epsilon^2). \tag{F.4}$$

Retaining only terms up to the order ϵ, we find that the perturbations satisfy:

$$\begin{aligned}\frac{\partial U(X,t)}{\partial t} &= \frac{1}{\mu^*}\frac{\partial}{\partial X}\left[\frac{d_1(u^*(t),v^*(t))}{\mu^*}\frac{\partial U(X,t)}{\partial X} + \frac{d_2(u^*(t),v^*(t))}{\mu^*}\frac{\partial V(X,t)}{\partial X}\right] \\ &+ \left[\frac{\partial f}{\partial u}(u^*,v^*) - u^*\frac{\partial S}{\partial u}(t,u^*,v^*) - S(t,u^*,v^*)\right]U(X,t) \\ &+ \left[\frac{\partial f}{\partial v}(u^*,v^*) - u^*\frac{\partial S}{\partial v}(t,u^*,v^*)\right]V(X,t).\end{aligned} \tag{F.5}$$

Let $H^* := H(u^*,v^*)$ where $H \in \{d_1,d_2,d_3,d_4,f,g\}$ and $S^* := S(t,u^*,v^*)$. We obtain:

$$\begin{aligned}\frac{\partial U(X,t)}{\partial t} &= \frac{1}{(\mu^*)^2}\frac{\partial}{\partial X}\left[d_1^*\frac{\partial U(X,t)}{\partial X} + d_2^*\frac{\partial V(X,t)}{\partial X}\right] + \left[\frac{\partial f^*}{\partial u} - u^*\frac{\partial S^*}{\partial u} - S^*\right]U(X,t) \\ &+ \left[\frac{\partial f^*}{\partial v} - u^*\frac{\partial S^*}{\partial v}\right]V(X,t).\end{aligned} \tag{F.6}$$

Similarly, we find:

$$\begin{aligned}\frac{\partial V(X,t)}{\partial t} &= \frac{1}{(\mu^*)^2}\frac{\partial}{\partial X}\left[d_3^*\frac{\partial U(X,t)}{\partial X} + d_4^*\frac{\partial V(X,t)}{\partial X}\right] + \left[\frac{\partial g^*}{\partial u} - v^*\frac{\partial S^*}{\partial u}\right]U(X,t) \\ &+ \left[\frac{\partial g^*}{\partial v} - v^*\frac{\partial S^*}{\partial v} - S^*\right]V(X,t).\end{aligned} \tag{F.7}$$

Moreover,

$$\begin{aligned}\frac{\partial \mu^*}{\partial t} + \epsilon\frac{\partial M}{\partial t} &= (\mu^*+\epsilon M)\left[S(t,u^*,v^*) + \epsilon\left(\frac{\partial S}{\partial u}(t,u^*,v^*)U + \frac{\partial S}{\partial v}(t,u^*,v^*)V\right)\right] \\ &= \mu^* S(t,u^*,v^*) + \mu^*\epsilon\left(\frac{\partial S}{\partial u}(t,u^*,v^*)U + \frac{\partial S}{\partial v}(t,u^*,v^*)V\right) + \epsilon M S(t,u^*,v^*)\end{aligned} \tag{F.8}$$

$$\Leftrightarrow \frac{\partial M}{\partial t} = \mu^*\left(\frac{\partial S}{\partial u}(t,u^*,v^*)U + \frac{\partial S}{\partial v}(t,u^*,v^*)V\right) + MS(t,u^*,v^*). \tag{F.9}$$

F.2 Matrix formulation

Let $\mathbf{W} = \begin{pmatrix} U \\ V \end{pmatrix}$, $\mathbf{D}^*(t) = \begin{pmatrix} d_1^*(t) & d_2^*(t) \\ d_3^*(t) & d_4^*(t) \end{pmatrix}$ and $\mathbf{A}(t) = \begin{pmatrix} \frac{\partial f^*}{\partial u} - u^* \frac{\partial S^*}{\partial u} - S^* & \frac{\partial f^*}{\partial v} - u^* \frac{\partial S^*}{\partial v} \\ \frac{\partial g^*}{\partial u} - v^* \frac{\partial S^*}{\partial u} & \frac{\partial g^*}{\partial v} - v^* \frac{\partial S^*}{\partial v} - S^* \end{pmatrix}$.

Then the system can be written in the following matrix form:

$$\frac{\partial \mathbf{W}(X,t)}{\partial t} = \frac{\mathbf{D}^*}{(\mu^*)^2} \frac{\partial^2 \mathbf{W}}{\partial X^2} + \mathbf{A}(t)\mathbf{W}. \tag{F.10}$$

F.3 Mode decomposition

To determine which spatial structures can grow, we decompose the perturbation $\mathbf{W}$ into elementary spatial modes. Because the domain is an interval with homogeneous Neumann boundary conditions, a natural basis is provided by the eigenfunctions of the Laplacian satisfying these boundary conditions.

$$\mathbf{W}(X,t) = \sum_{k=0}^{+\infty} \psi_k(X) \tilde{\mathbf{W}}_k(t), \tag{F.11}$$

$$\frac{\partial^2 \psi_k}{\partial X^2} = -\rho_k \psi_k(X), \tag{F.12}$$

where the constant spatial eigenvalues $\rho_k = \left(\frac{k\pi}{L_0}\right)^2$ correspond to the standard Laplacian on an interval $[0, L_0]$ with associated eigenfunctions $\psi_k(X) = \cos\left(\frac{k\pi X}{L_0}\right)$.

Inserting this expression of $\mathbf{W}$ into (F.10), we find:

$$\begin{aligned} \sum_{k=0}^{+\infty} \psi_k(X) \frac{\partial \tilde{\mathbf{W}}_\mathbf{k}(t)}{\partial t} &= \frac{\mathbf{D}^*}{(\mu^*)^2} \sum_{k=0}^{+\infty} \frac{\partial^2 \psi_k(X)}{\partial X^2} \tilde{\mathbf{W}}_k(t) + \mathbf{A}(t) \sum_{k=0}^{+\infty} \psi_k(X) \tilde{\mathbf{W}}_k(t) \\ &= \sum_{k=0}^{+\infty} \left[\left(-\frac{\mathbf{D}^*(t)}{(\mu^*(t))^2} \rho_k + \mathbf{A}(t) \right) \psi_k(X) \tilde{\mathbf{W}}_k(t) \right]. \end{aligned} \tag{F.13}$$

By uniqueness of the decomposition in the eigenfunction basis ψ_k, we obtain:

$$\frac{\partial \tilde{\mathbf{W}}_k(t)}{\partial t} = \left(-\frac{\mathbf{D}^*(t)}{(\mu^*(t))^2} \rho_k + \mathbf{A}(t) \right) \tilde{\mathbf{W}}_k(t). \tag{F.14}$$

Writing $\mathbf{B}_k(t) = \begin{pmatrix} b_{11}(t) & b_{12}(t) \\ b_{21}(t) & b_{22}(t) \end{pmatrix} = -\frac{\mathbf{D}^*(t)}{(\mu^*(t))^2}\rho_k + \mathbf{A}(t)$, equation (F.14) becomes:

$$\frac{\partial \tilde{\mathbf{W}}_k(t)}{\partial t} = \mathbf{B}_k(t)\tilde{\mathbf{W}}_k(t). \tag{F.15}$$

Appendix G

Details of the calculations - Reduction to a third-order differential equation for a three-species system

For each k = 0, 1, 2, ..., equation (3.37) can be written as the system:

$$\frac{d\tilde{U}_k(t)}{dt} = b_{11}(t)\tilde{U}_k(t) + b_{12}(t)\tilde{V}_k(t) + b_{13}(t)\tilde{P}_k(t), \tag{G.1}$$

$$\frac{d\tilde{V}_k(t)}{dt} = b_{21}(t)\tilde{U}_k(t) + b_{22}(t)\tilde{V}_k(t) + b_{23}(t)\tilde{P}_k(t), \tag{G.2}$$

$$\frac{d\tilde{P}_k(t)}{dt} = b_{31}(t)\tilde{U}_k(t) + b_{32}(t)\tilde{V}_k(t) + b_{33}(t)\tilde{P}_k(t). \tag{G.3}$$

We derive equation (G.1) and obtain:

$$\tilde{U}_k''(t) = \beta_1\tilde{U}_k(t) + \beta_2\tilde{V}_k(t) + \beta_3\tilde{P}_k(t), \tag{G.4}$$

where

$$\beta_1 = b_{11}^2(t) + b_{12}(t)b_{21}(t) + b_{13}(t)b_{31}(t) + b_{11}'(t), \tag{G.5}$$

$$\beta_2 = b_{11}(t)b_{12}(t) + b_{12}(t)b_{22}(t) + b_{13}(t)b_{32}(t) + b_{12}'(t), \tag{G.6}$$

and

$$\beta_3 = b_{11}(t)b_{13}(t) + b_{12}(t)b_{23}(t) + b_{13}(t)b_{33}(t) + b_{13}'(t). \tag{G.7}$$

From equation (G.1), we have:

$$\tilde{U}_k'(t) - b_{11}(t)\tilde{U}_k(t) = b_{12}(t)\tilde{V}_k(t) + b_{13}(t)\tilde{P}_k(t). \tag{G.8}$$

On the other hand, equation (G.4) gives:

$$\tilde{U}_k''(t) - \beta_1\tilde{U}_k(t) = \beta_2\tilde{V}_k(t) + \beta_3\tilde{P}_k(t). \tag{G.9}$$

Then:

$$\begin{pmatrix} \tilde{U}_k'(t) - b_{11}(t)\tilde{U}_k(t) \\ \tilde{U}_k''(t) - \beta_1\tilde{U}_k(t) \end{pmatrix} = \begin{pmatrix} b_{12}(t) & b_{13}(t) \\ \beta_2 & \beta_3 \end{pmatrix} \begin{pmatrix} \tilde{V}_k(t) \\ \tilde{P}_k(t) \end{pmatrix}. \tag{G.10}$$

Let $\Delta = \beta_3 b_{12}(t) - \beta_2 b_{13}(t)$. Under the assumption that $\Delta \neq 0$, we obtain:

$$\tilde{V}_k(t) = \frac{\beta_3(\tilde{U}_k'(t) - b_{11}(t)\tilde{U}_k(t)) - b_{13}(t)(\tilde{U}_k''(t) - \beta_1\tilde{U}_k(t))}{\Delta}, \tag{G.11}$$

$$\tilde{P}_k(t) = \frac{b_{12}(t)(\tilde{U}_k''(t) - \beta_1\tilde{U}_k(t)) - \beta_2(\tilde{U}_k'(t) - b_{11}(t)\tilde{U}_k(t))}{\Delta}. \tag{G.12}$$

Then, deriving equation (G.4), we obtain:

$$\tilde{U}_k'''(t) = \gamma_1\tilde{U}_k(t) + \gamma_2\tilde{V}_k(t) + \gamma_3\tilde{P}_k(t), \tag{G.13}$$

where

$$\gamma_1 = \beta_1 b_{11}(t) + \beta_2 b_{21}(t) + \beta_3 b_{31}(t) + \beta_1', \tag{G.14}$$

$$\gamma_2 = \beta_1 b_{12}(t) + \beta_2 b_{22}(t) + \beta_3 b_{32}(t) + \beta_2', \tag{G.15}$$

and

$$\gamma_3 = \beta_1 b_{13}(t) + \beta_2 b_{23}(t) + \beta_3 b_{33}(t) + \beta_3'. \tag{G.16}$$

Using equations (G.11) and (G.12), we obtain:

$$\tilde{U}_k'''(t) + P_{U_k}(t)\tilde{U}_k''(t) + Q_{U_k}(t)\tilde{U}_k'(t) + R_{U_k}(t)\tilde{U}_k(t) = 0, \tag{G.17}$$

with:

$$P_{U_k}(t) = \frac{\gamma_2}{\Delta}b_{13} - \frac{\gamma_3}{\Delta}b_{12}, \tag{G.18}$$

$$Q_{U_k}(t) = \frac{\gamma_3}{\Delta}\beta_2 - \frac{\gamma_2}{\Delta}\beta_3, \tag{G.19}$$

and:

$$R_{U_k}(t) = -\gamma_1 + \frac{\gamma_2}{\Delta}(\beta_3 b_{11} - b_{13}\beta_1) + \frac{\gamma_3}{\Delta}(b_{12}\beta_1 - \beta_2 b_{11}). \tag{G.20}$$

Bibliography

[1] Alan M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 237(641):37–72, 1952.

[2] Frits Veerman, Moritz Mercker, and Anna Marciniak-Czochra. Beyond turing: Far-from-equilibrium patterns and mechano-chemical feedback. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 379:20200278, 2021.

[3] Edmund J. Crampin, Eamonn A. Gaffney, and Philip K. Maini. Reaction and diffusion on growing domains: Scenarios for robust pattern formation. *Bulletin of Mathematical Biology*, 61:1093–1120, 1999.

[4] Andrew L. Krause, Eamonn A. Gaffney, and Benjamin J. Walker. Concentration-dependent domain evolution in reaction-diffusion systems. *Bulletin of Mathematical Biology*, 85(14), 2023.

[5] Chiara Villa, Mark A. J. Chaplain, Alf Gerisch, and Tommaso Lorenzi. Mechanical models of pattern and form in biological tissues: The role of stress-strain constitutive equations. *Bulletin of Mathematical Biology*, 83(80), 2021.

[6] Alain Goriely. *The Mathematics and Mechanics of Biological Growth*, volume 45 of *Interdisciplinary Applied Mathematics*. Springer, 2017.

[7] Robert A. Van Gorder. Turing and benjamin-feir instability mechanisms in non-autonomous systems. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 476:20200003, 2020.

[8] J. D. Murray. *Mathematical Biology II: Spatial Models and Biomedical Applications*, volume 18 of *Interdisciplinary Applied Mathematics*. Springer, 3 edition, 2003.

[9] Robert A. Van Gorder, Václav Klika, and Andrew L. Krause. Turing conditions for pattern forming systems on evolving manifolds. *Journal of Mathematical Biology*, 82(1):4, 2021.

[10] Tadeusz Kaczorek. Externally and internally positive time-varying linear systems. In *15th Triennial World Congress*, Barcelona, Spain, 2002. IFAC.